

BFS Traversal

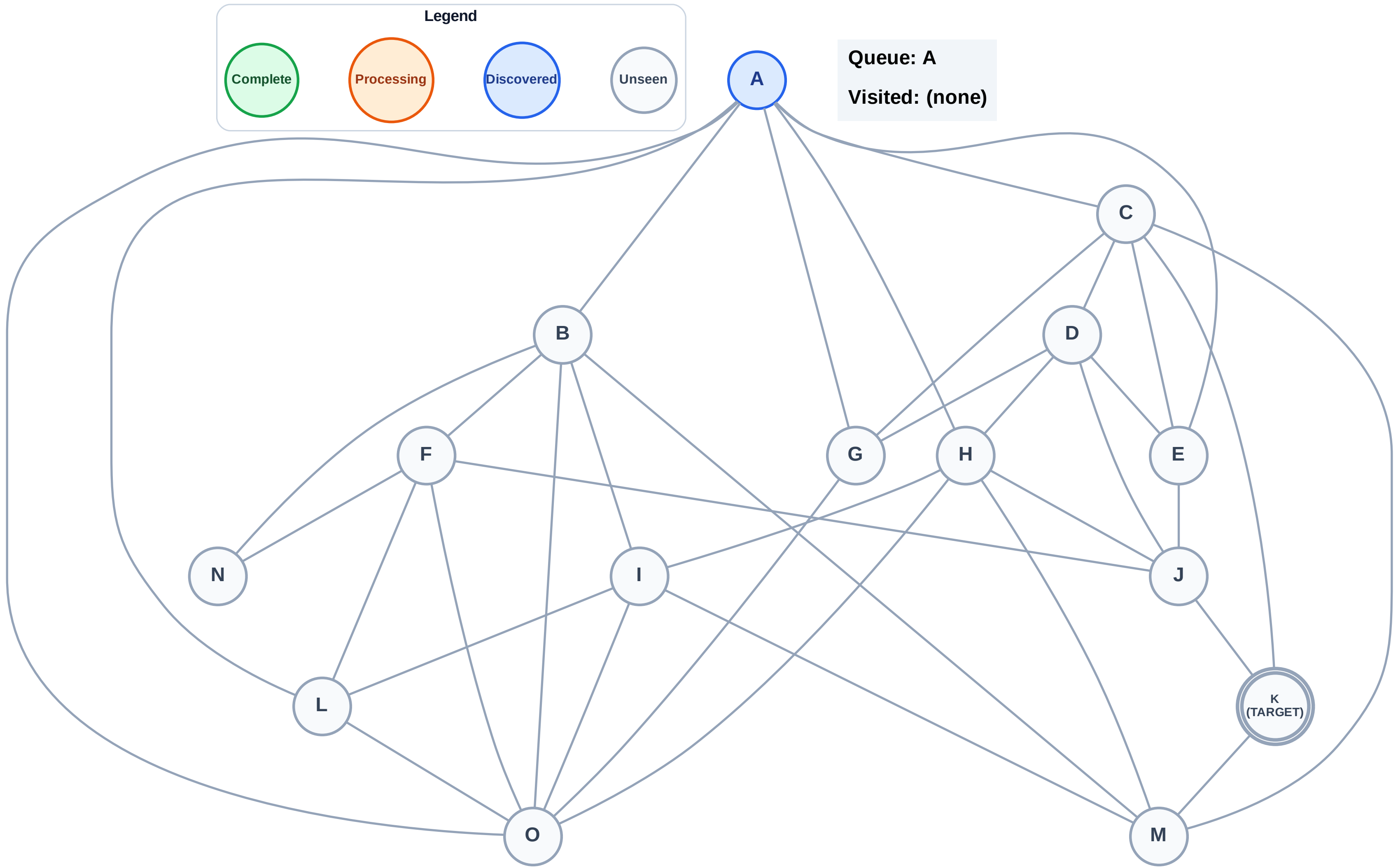
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

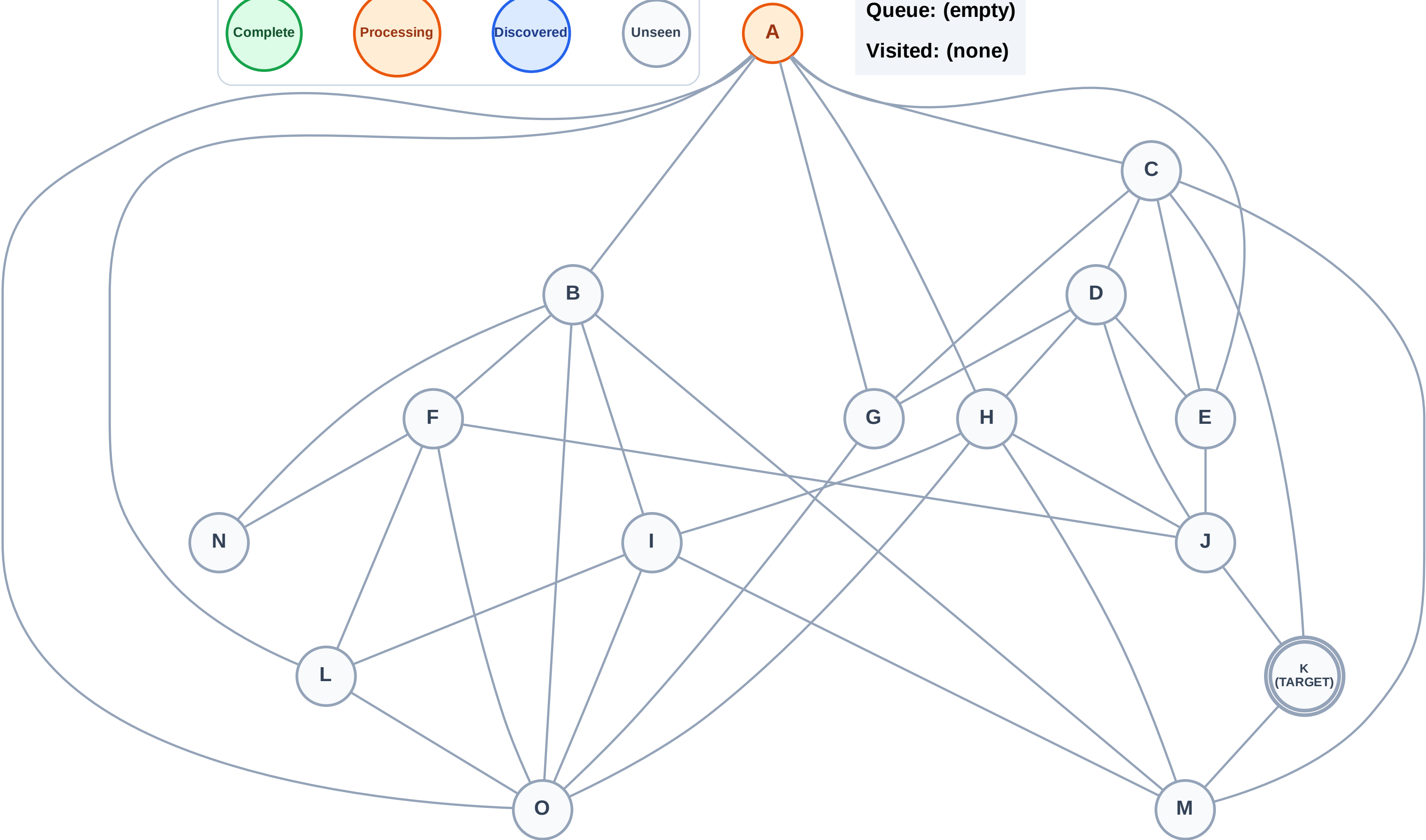
Complete

Processing

Discovered

Unseen

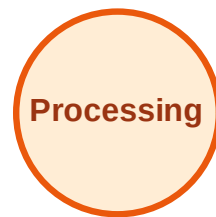
Queue: (empty)
Visited: (none)



BFS Traversal

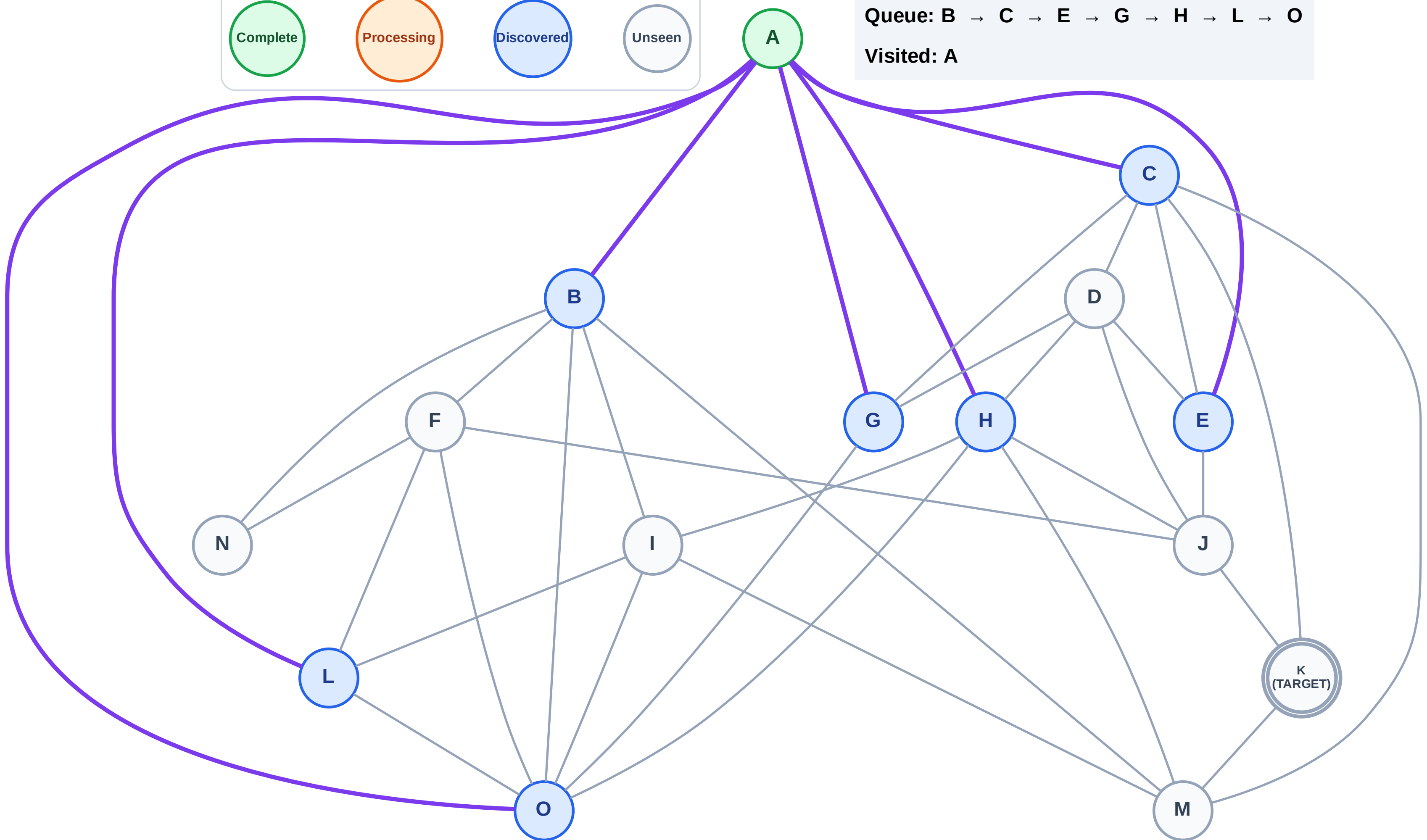
Step 3: Discover B, C, E, G, H, L, O from A

Legend



Queue: B → C → E → G → H → L → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

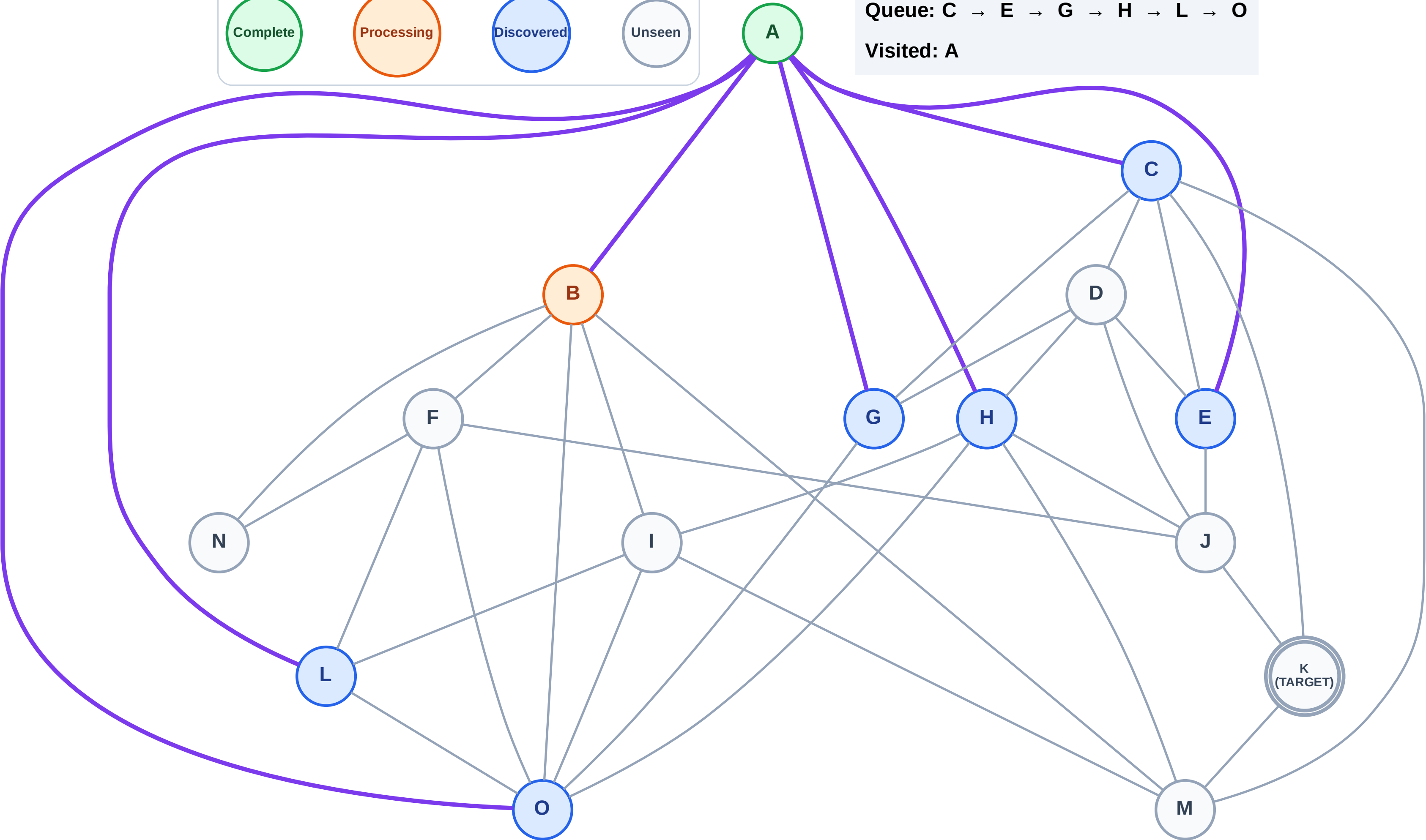
Processing

Discovered

Unseen

Queue: C → E → G → H → L → O

Visited: A



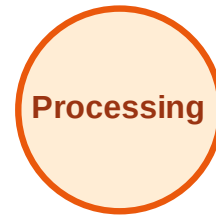
BFS Traversal

Step 5: Discover F, I, M, N from B

Legend



Complete



Processing



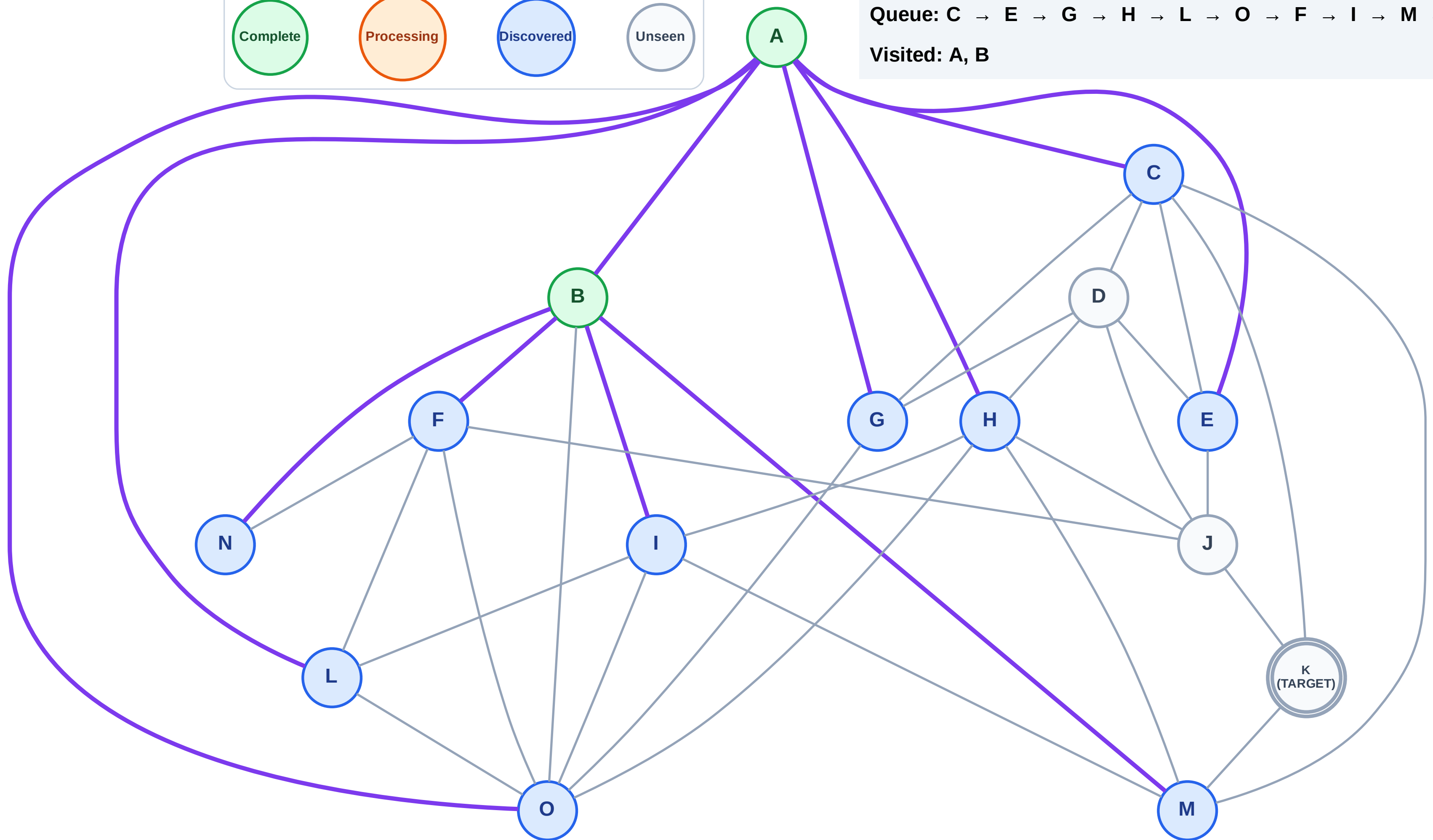
Discovered



Unseen

Queue: C → E → G → H → L → O → F → I → M → N

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

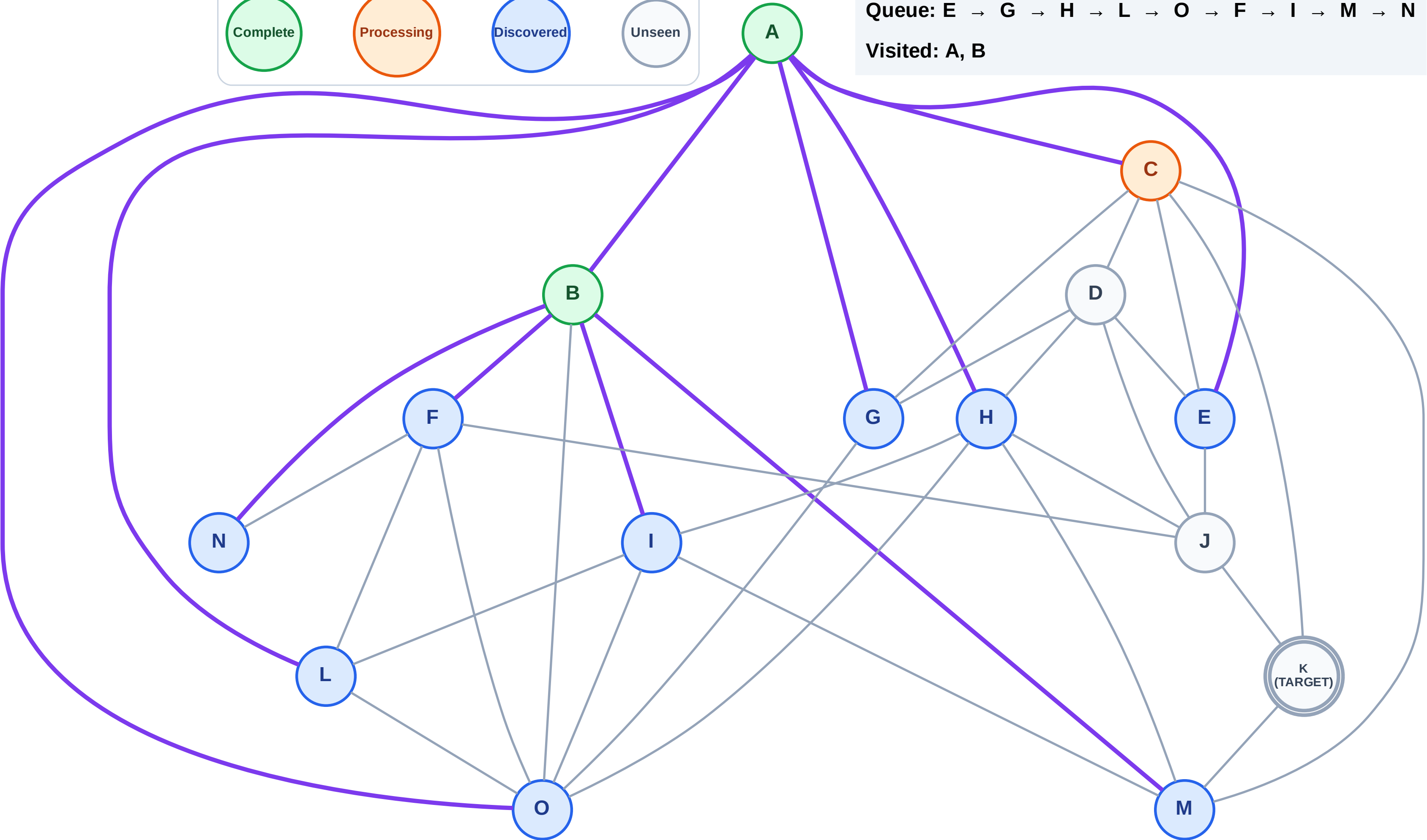
Processing

Discovered

Unseen

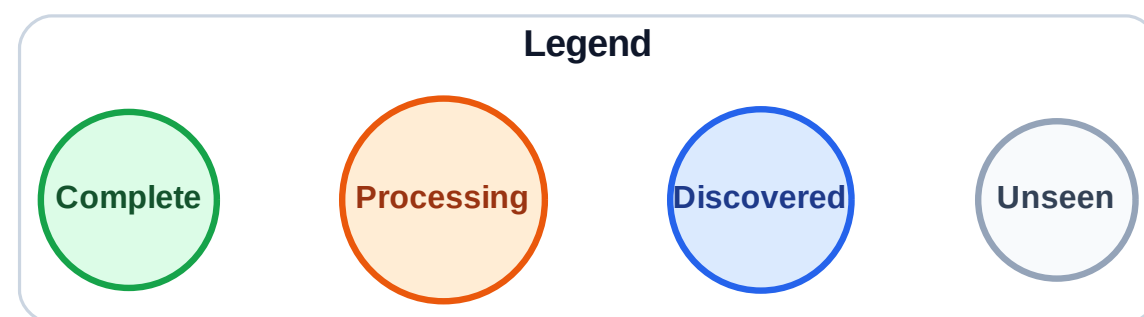
Queue: E → G → H → L → O → F → I → M → N

Visited: A, B



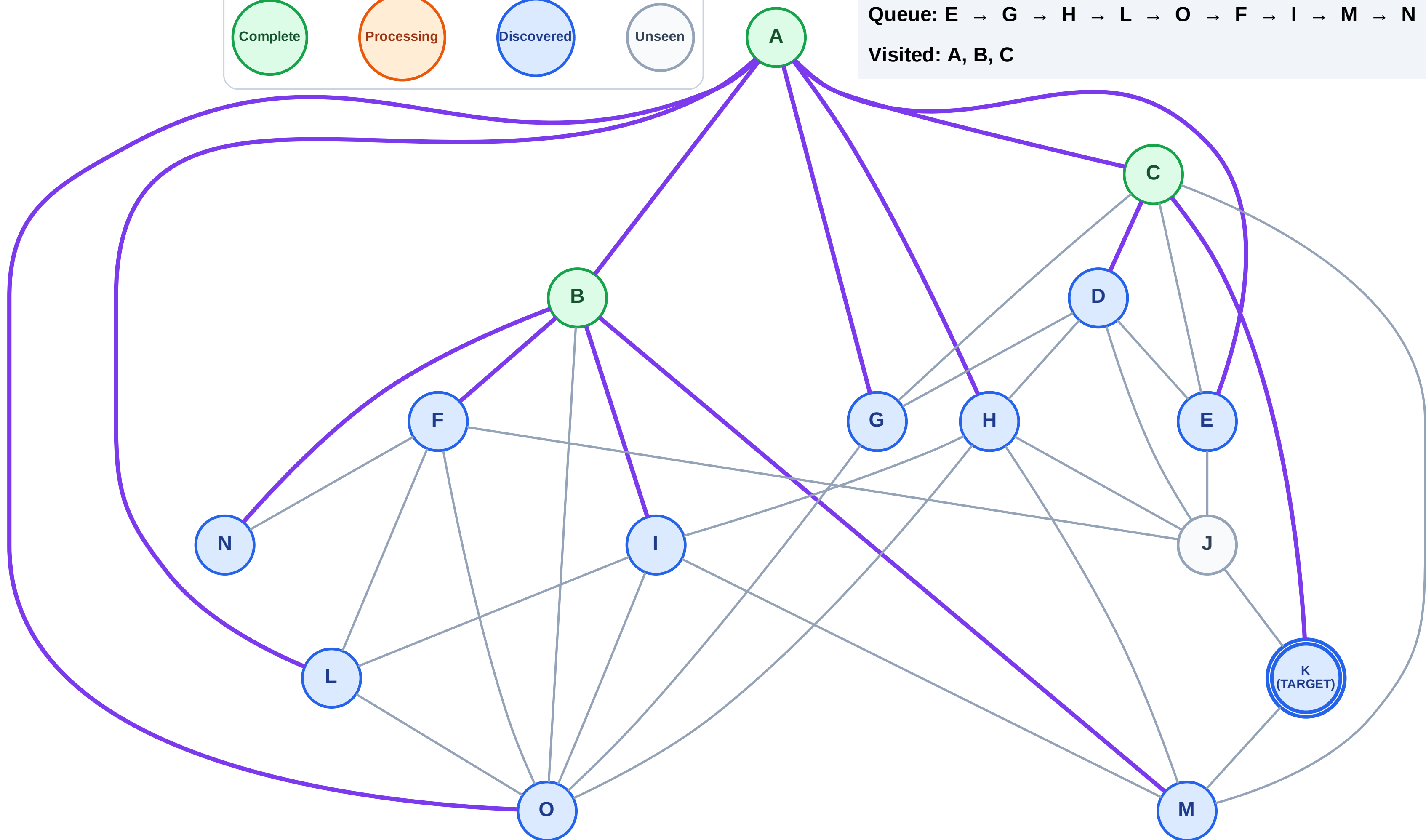
BFS Traversal

Step 7: Discover D, K from C



Queue: E → G → H → L → O → F → I → M → N → D → K

Visited: A, B, C



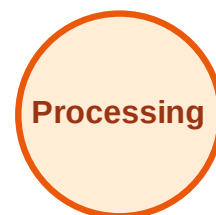
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



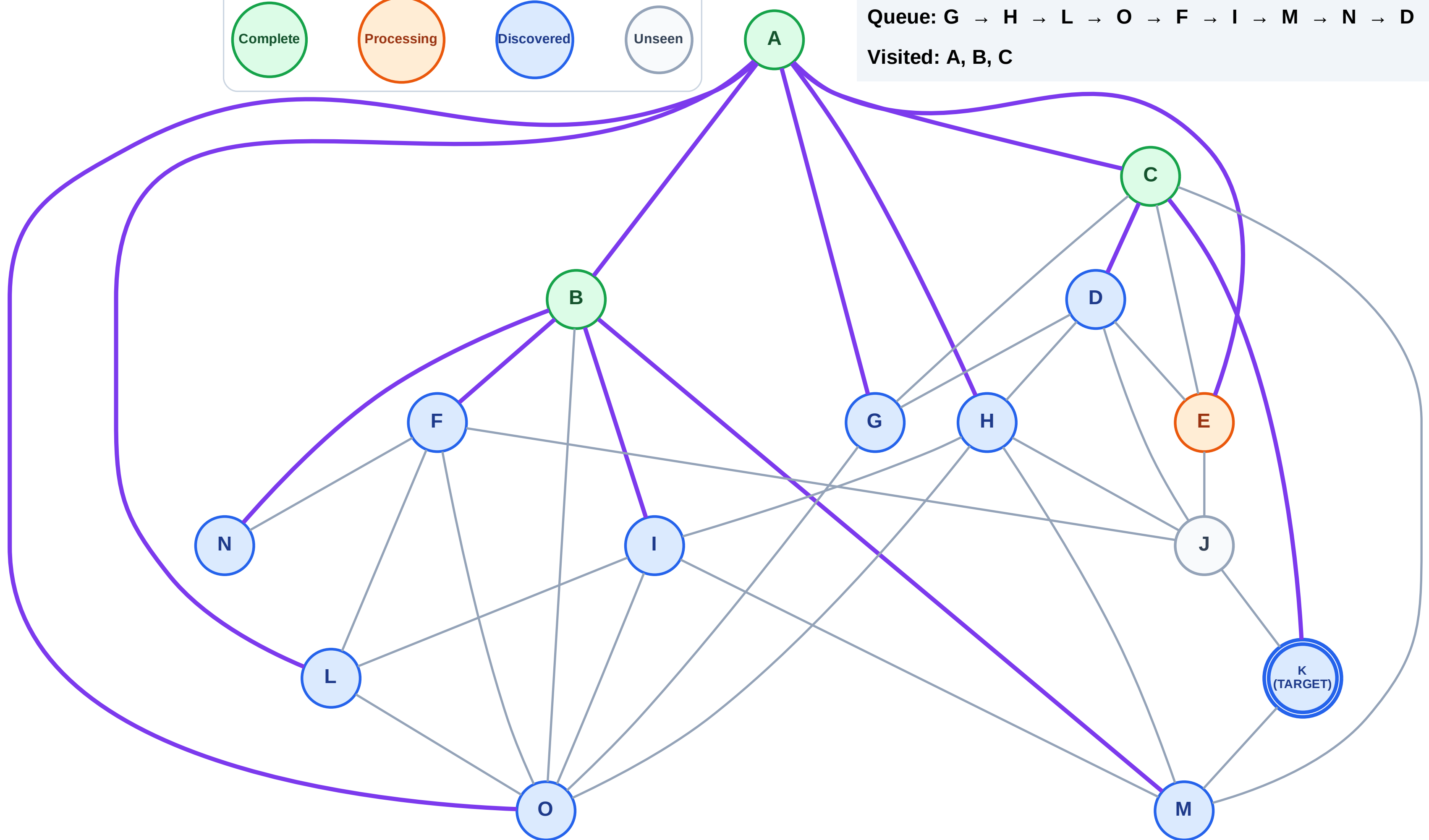
Discovered



Unseen

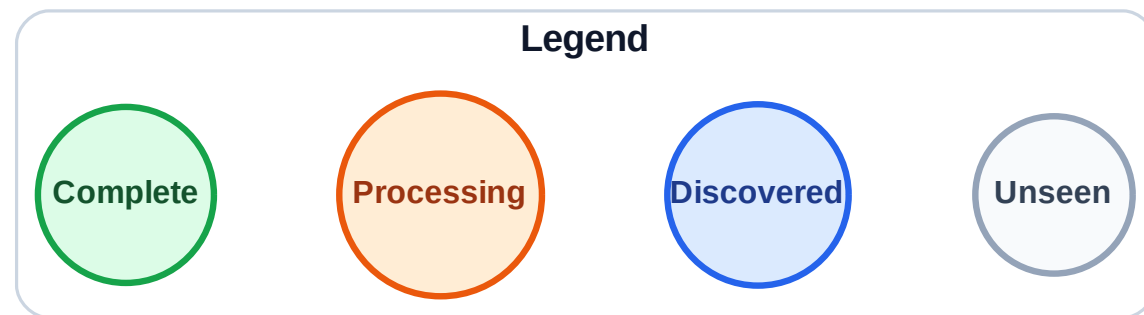
Queue: G → H → L → O → F → I → M → N → D → K

Visited: A, B, C



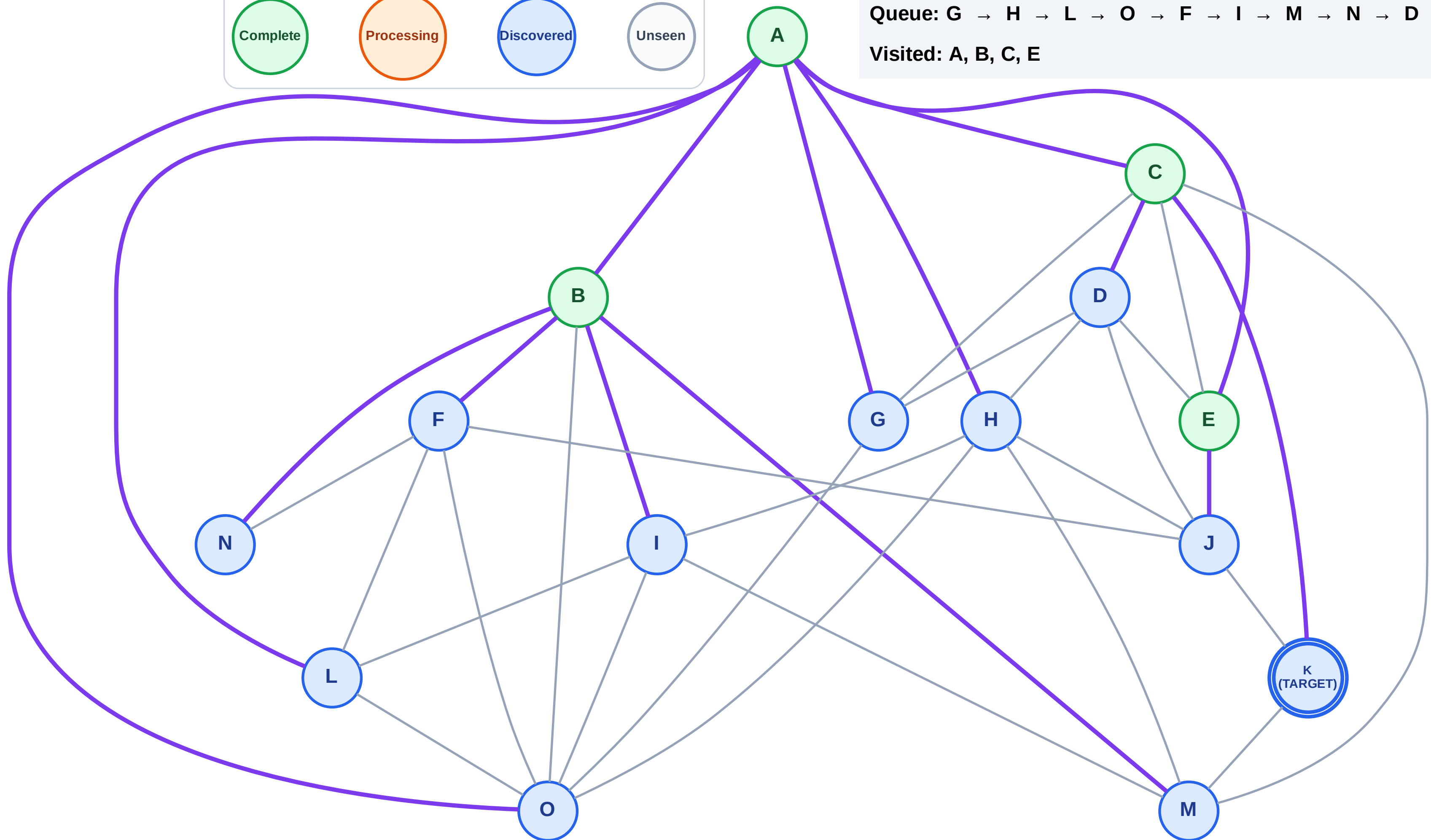
BFS Traversal

Step 9: Discover J from E



Queue: G → H → L → O → F → I → M → N → D → K → J

Visited: A, B, C, E



BFS Traversal

Step 10: Process node G

Legend

Complete

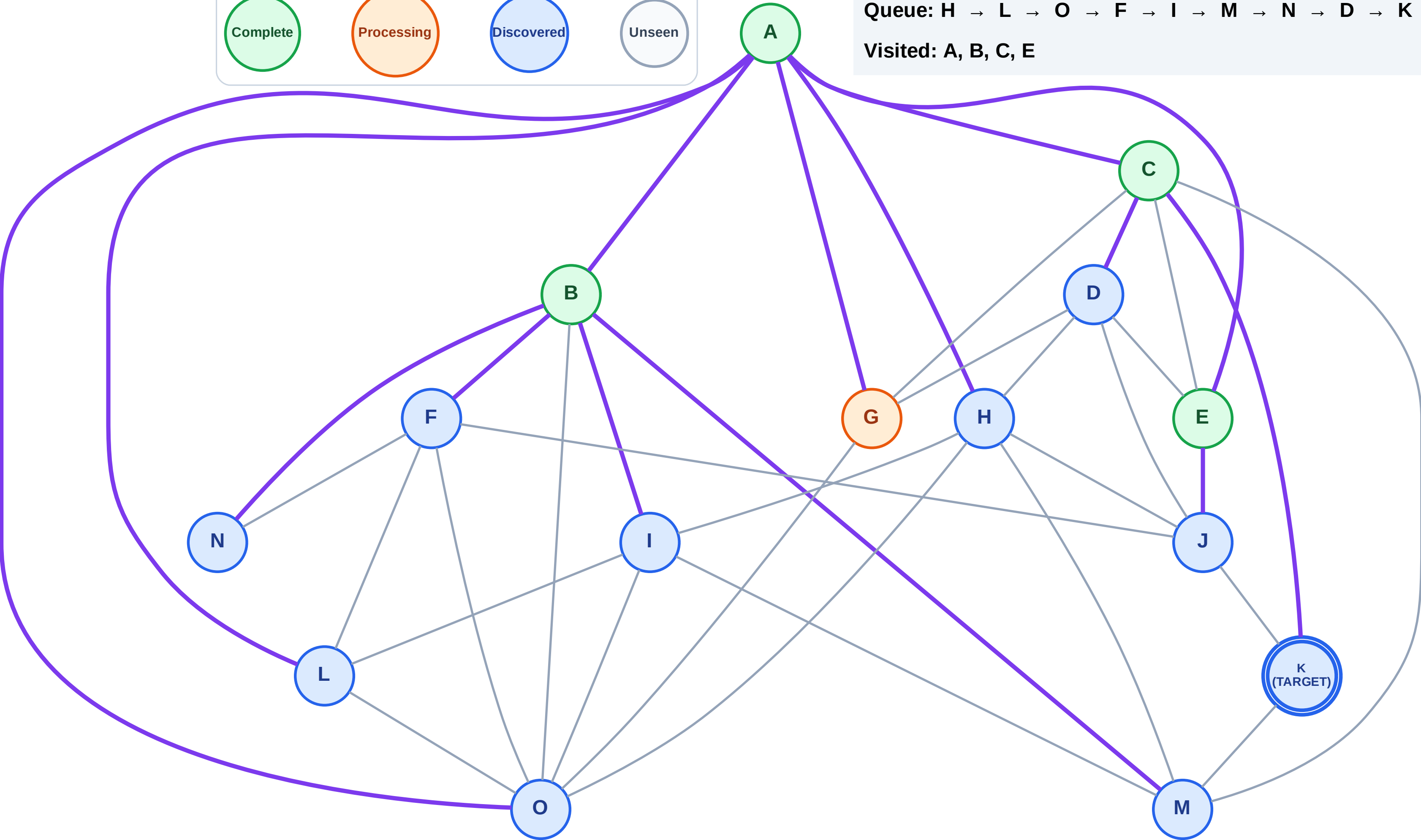
Processing

Discovered

Unseen

Queue: H → L → O → F → I → M → N → D → K → J

Visited: A, B, C, E



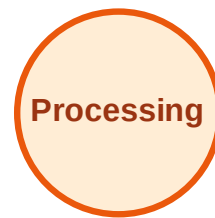
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



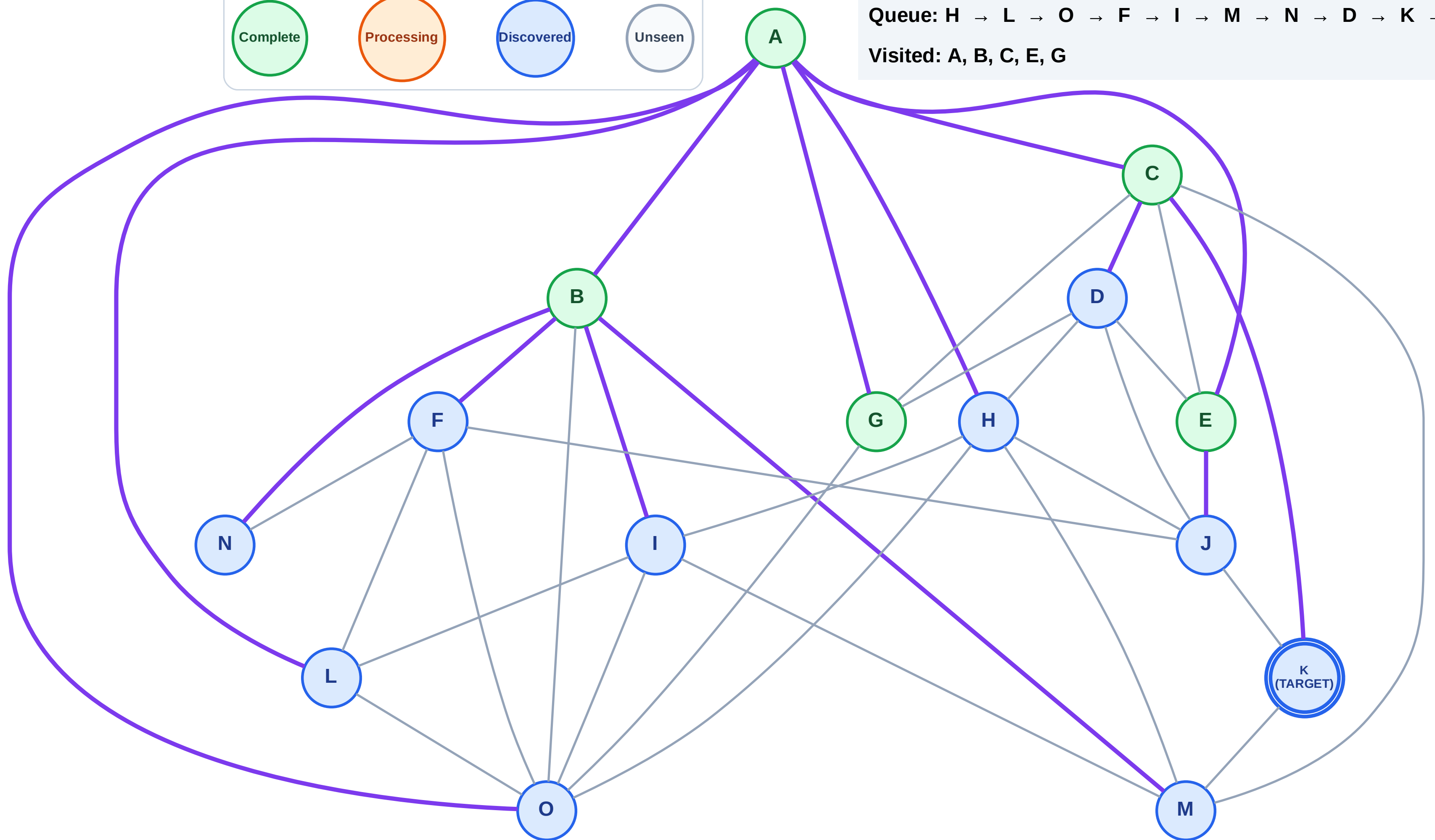
Discovered



Unseen

Queue: H → L → O → F → I → M → N → D → K → J

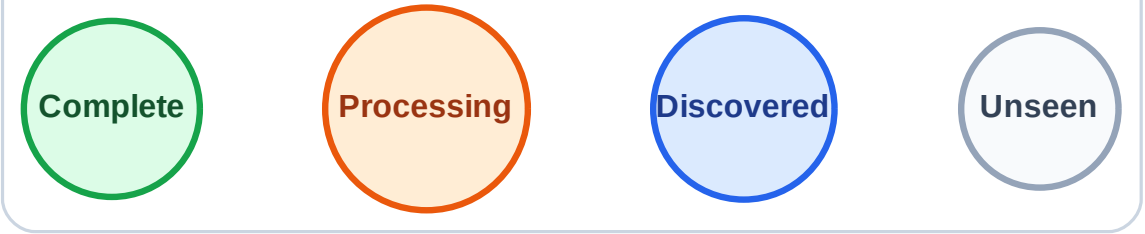
Visited: A, B, C, E, G



BFS Traversal

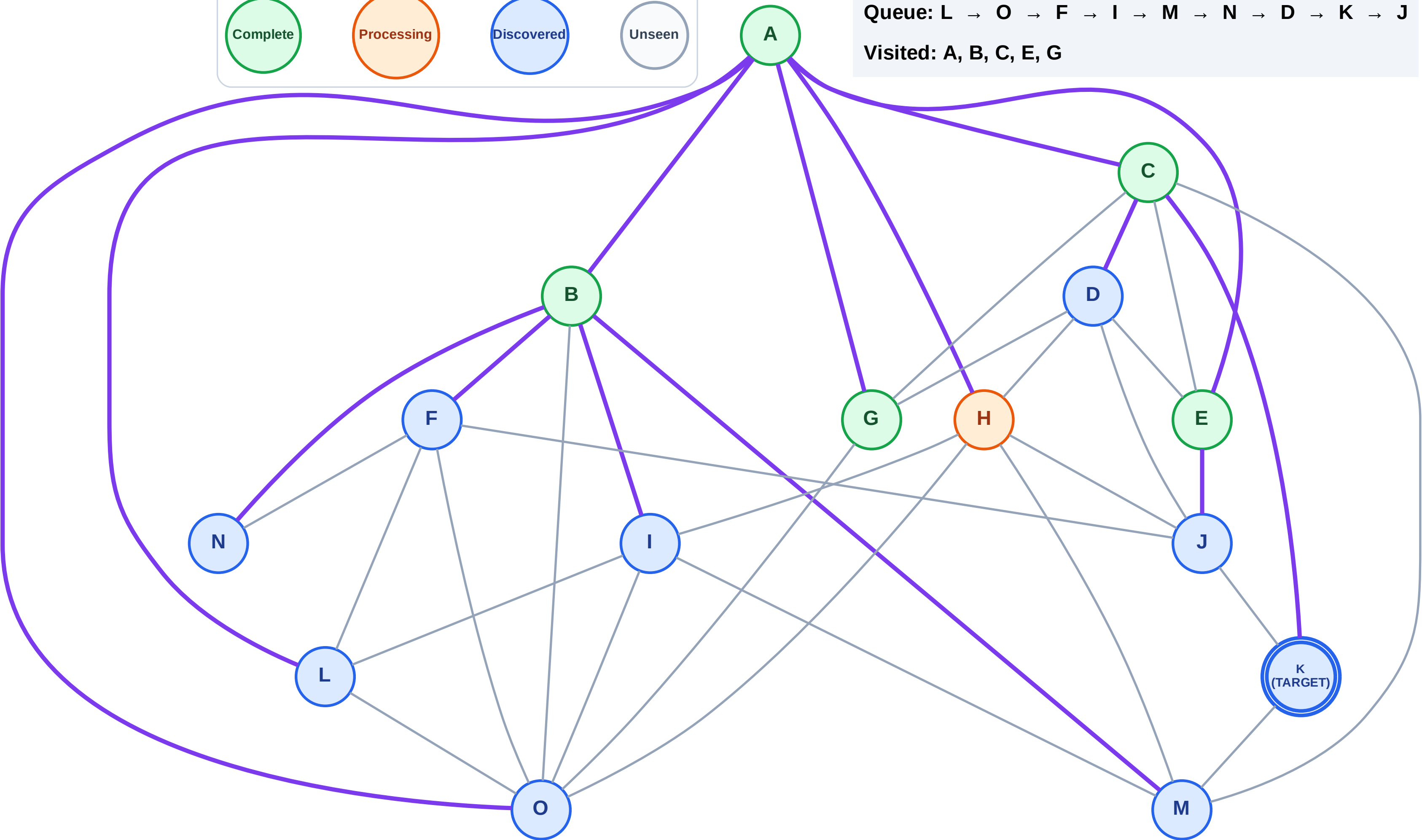
Step 12: Process node H

Legend



Queue: L → O → F → I → M → N → D → K → J

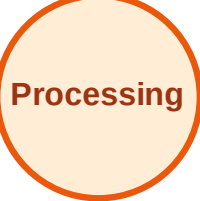
Visited: A, B, C, E, G



BFS Traversal

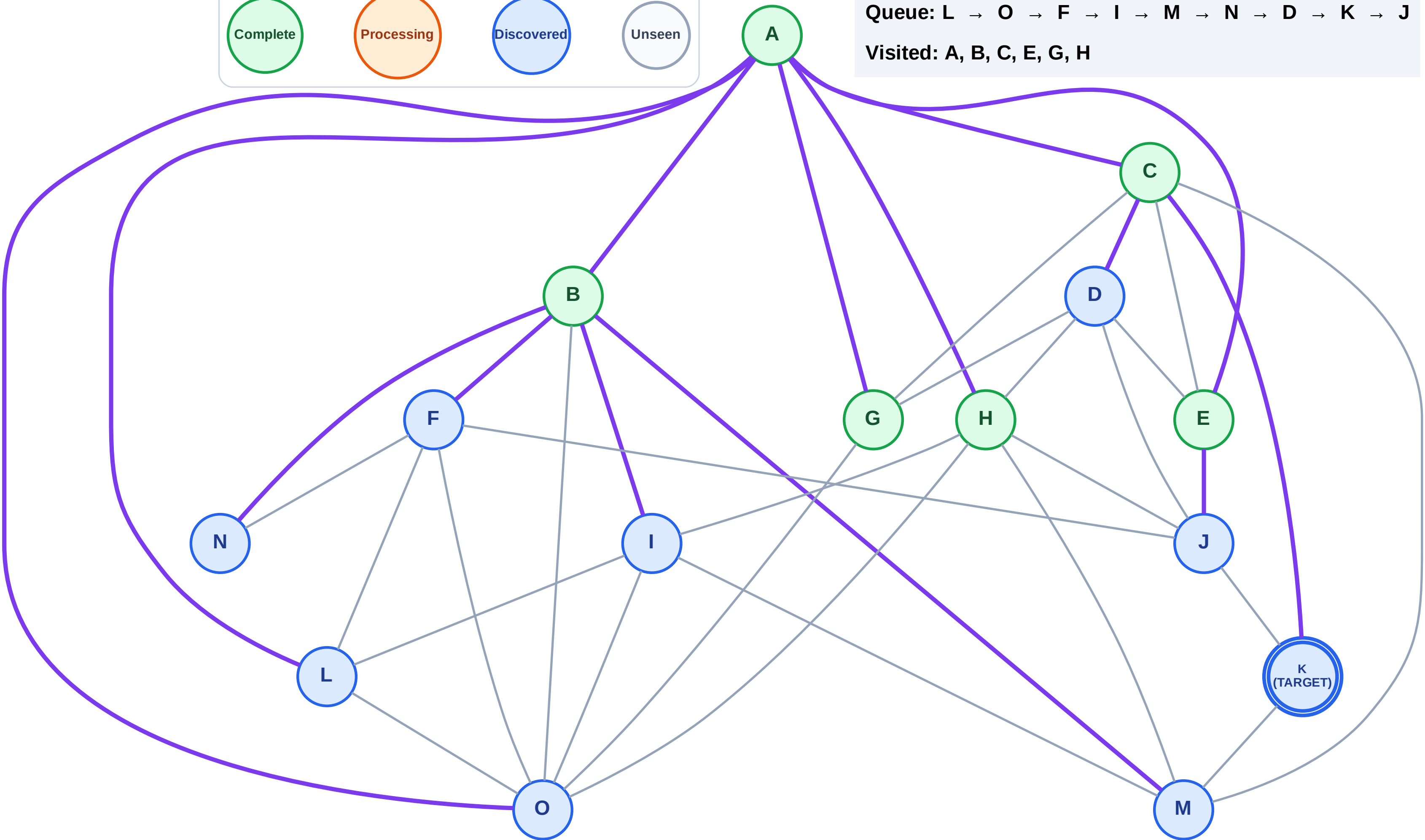
Step 13: No new neighbors from H

Legend



Queue: L → O → F → I → M → N → D → K → J

Visited: A, B, C, E, G, H



BFS Traversal

Step 14: Process node L

Legend

Complete

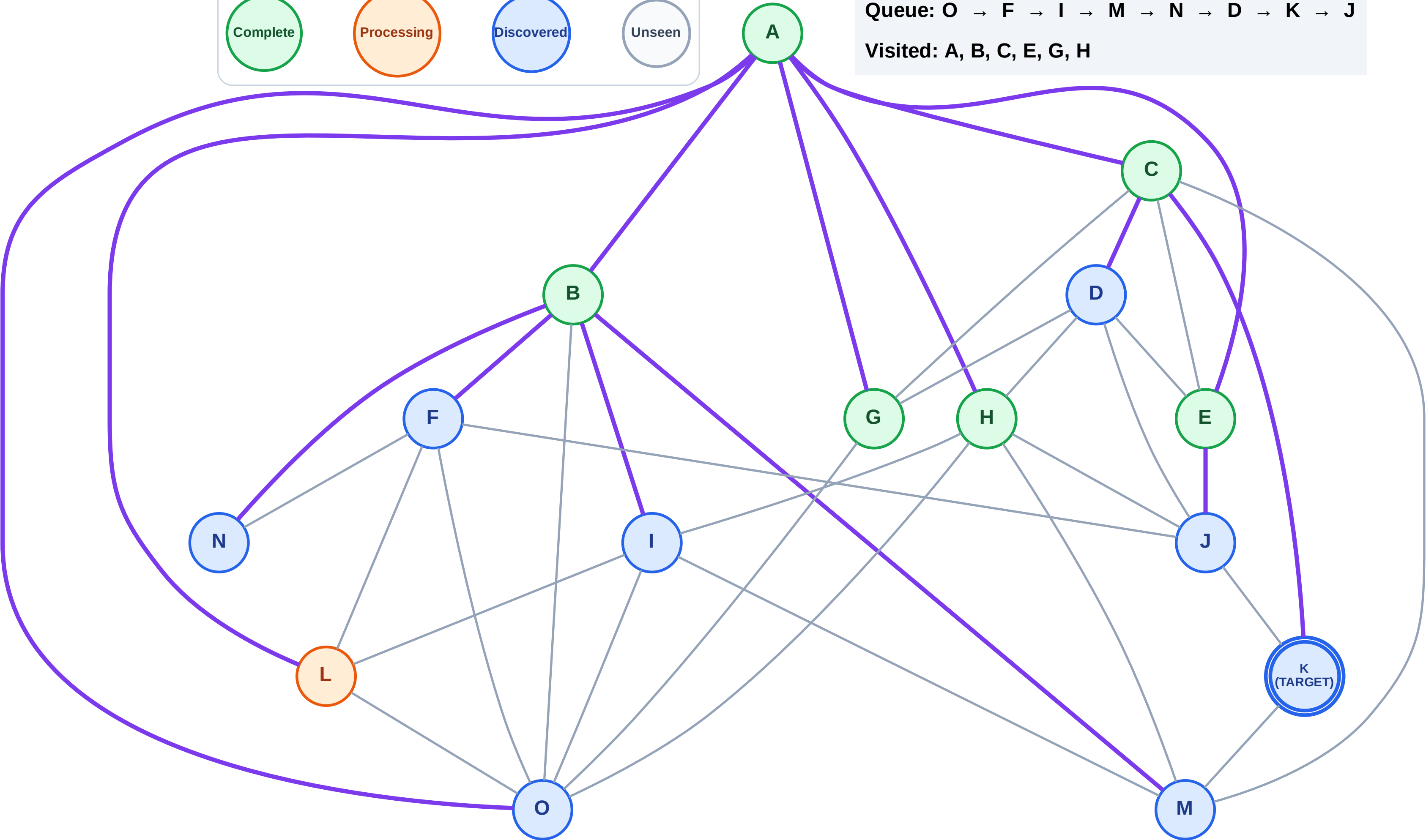
Processing

Discovered

Unseen

Queue: O → F → I → M → N → D → K → J

Visited: A, B, C, E, G, H



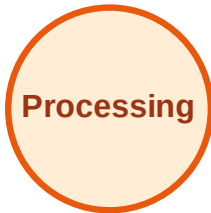
BFS Traversal

Step 15: No new neighbors from L

Legend



Complete



Processing



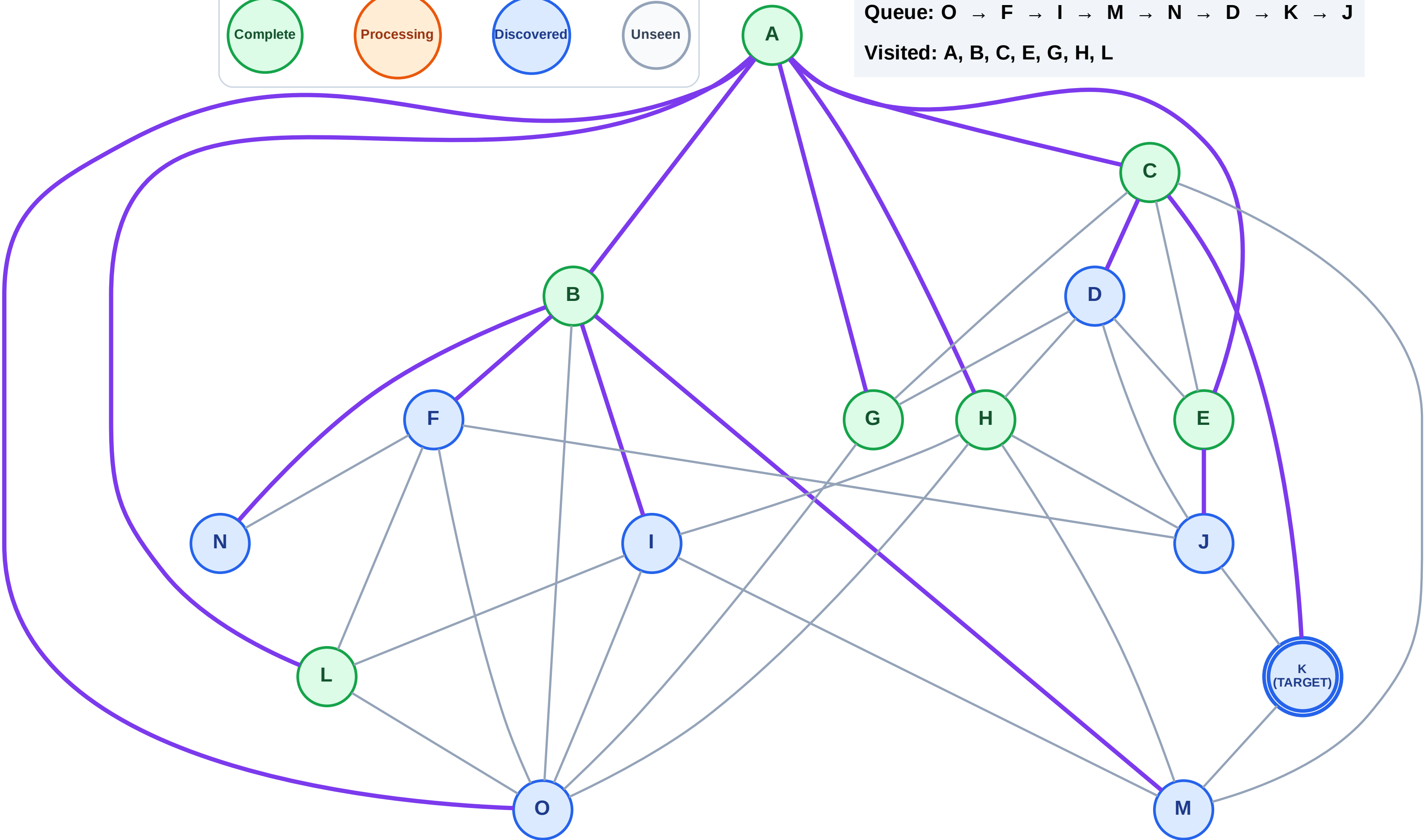
Discovered



Unseen

Queue: O → F → I → M → N → D → K → J

Visited: A, B, C, E, G, H, L



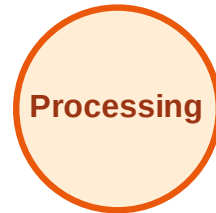
BFS Traversal

Step 16: Process node O

Legend



Complete



Processing



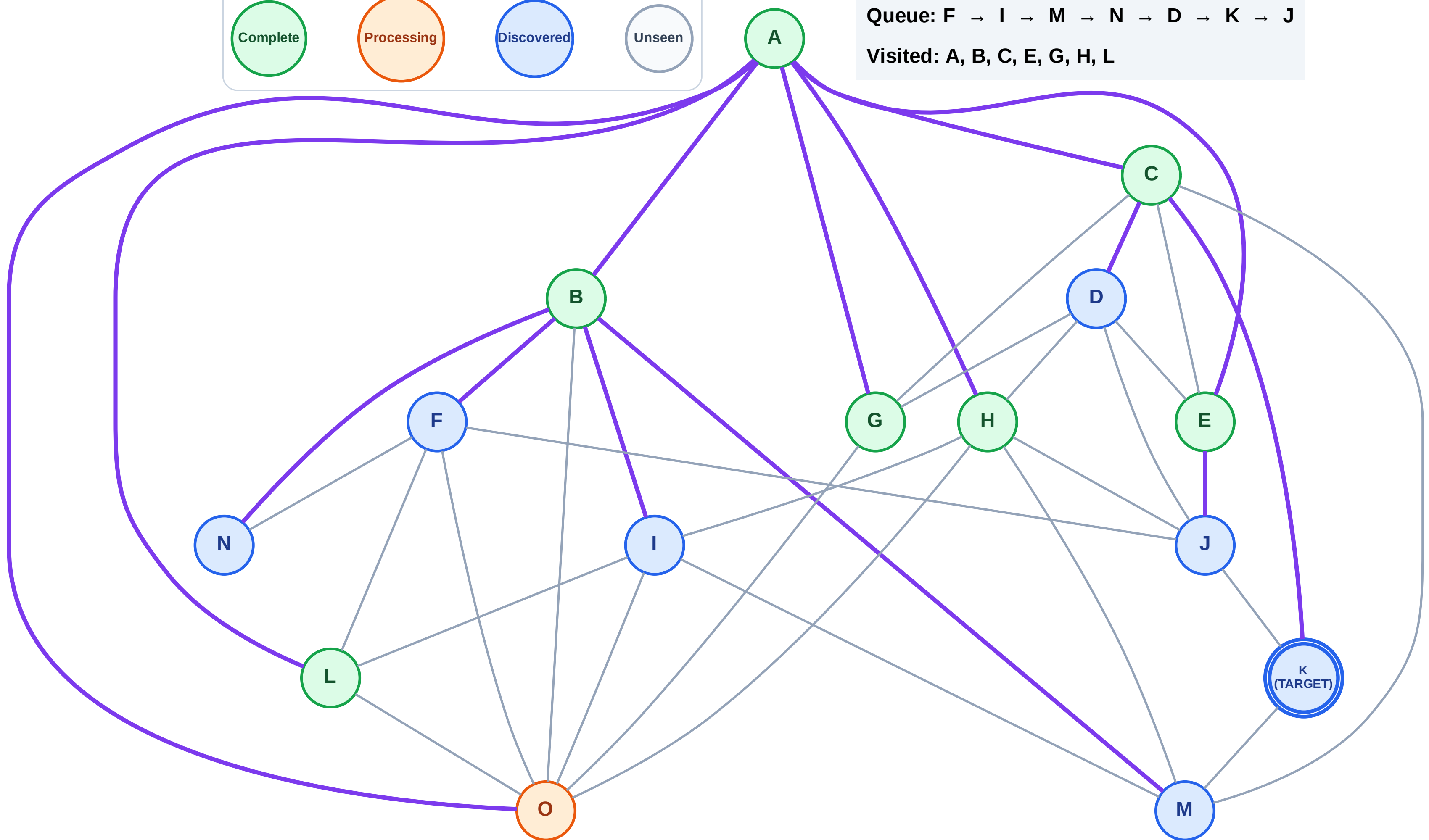
Discovered



Unseen

Queue: F → I → M → N → D → K → J

Visited: A, B, C, E, G, H, L



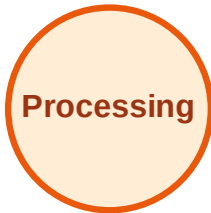
BFS Traversal

Step 17: No new neighbors from O

Legend



Complete



Processing



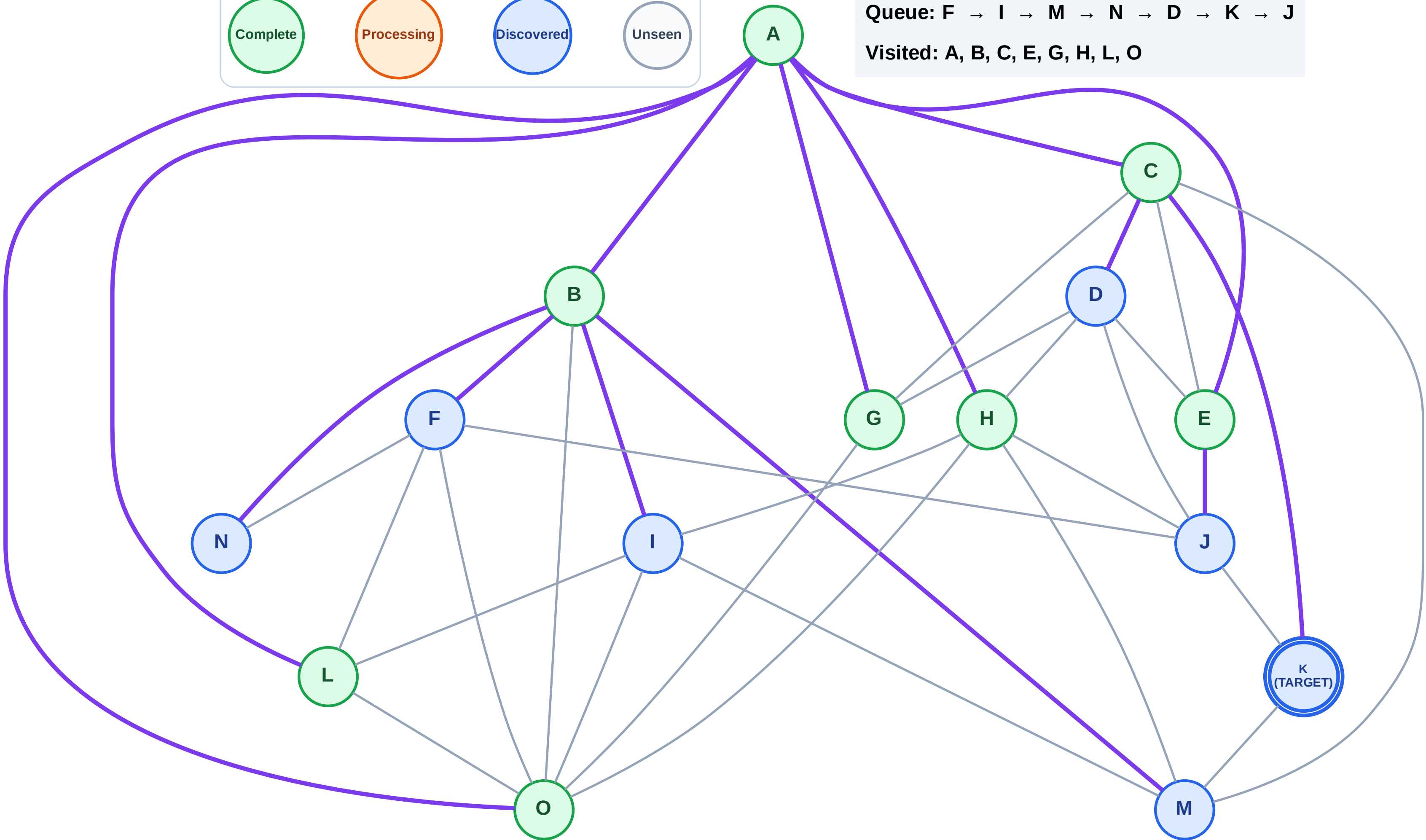
Discovered



Unseen

Queue: F → I → M → N → D → K → J

Visited: A, B, C, E, G, H, L, O



BFS Traversal

Step 18: Process node F

Legend

Complete

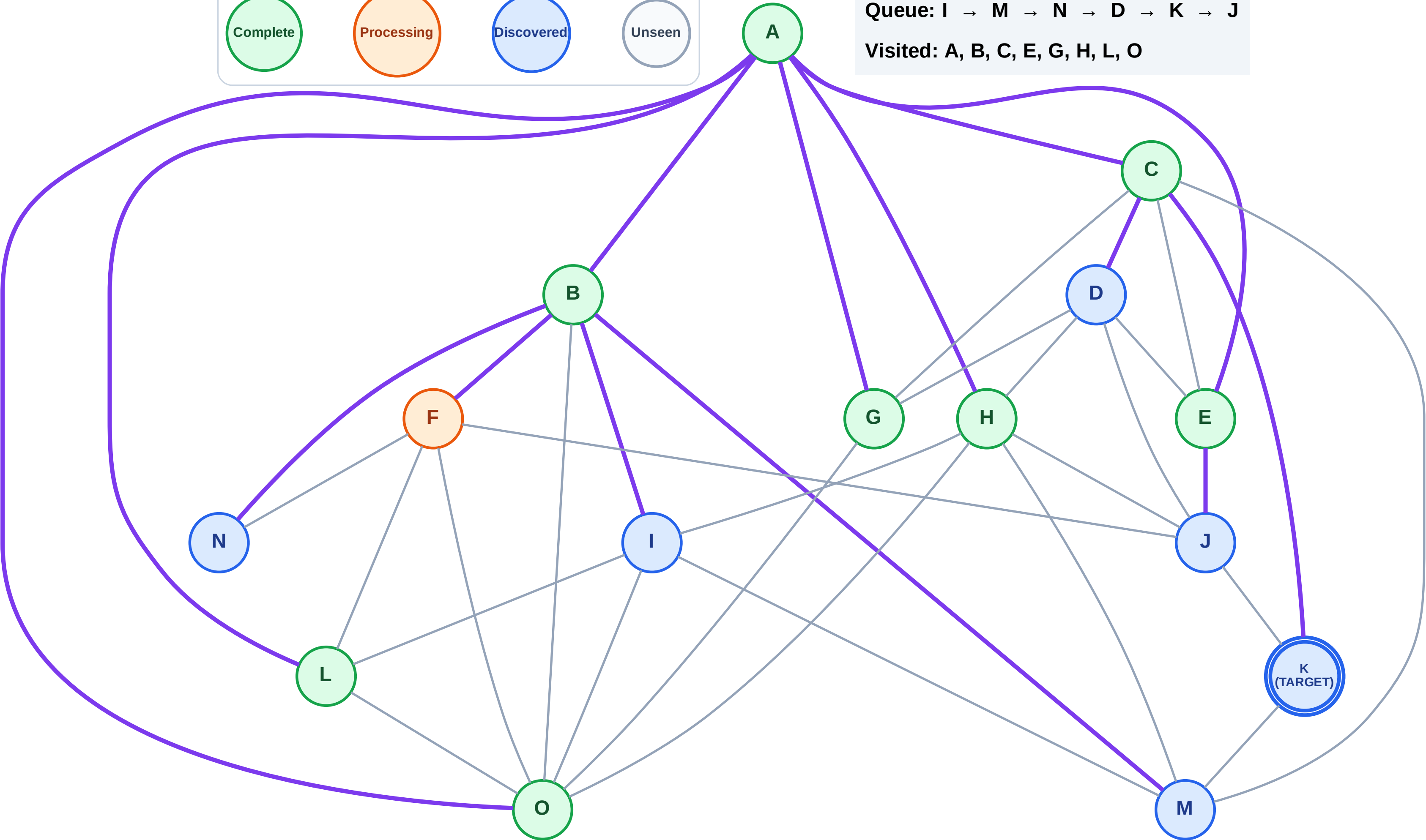
Processing

Discovered

Unseen

Queue: I → M → N → D → K → J

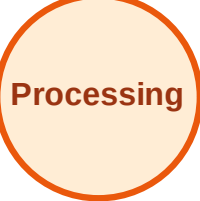
Visited: A, B, C, E, G, H, L, O



BFS Traversal

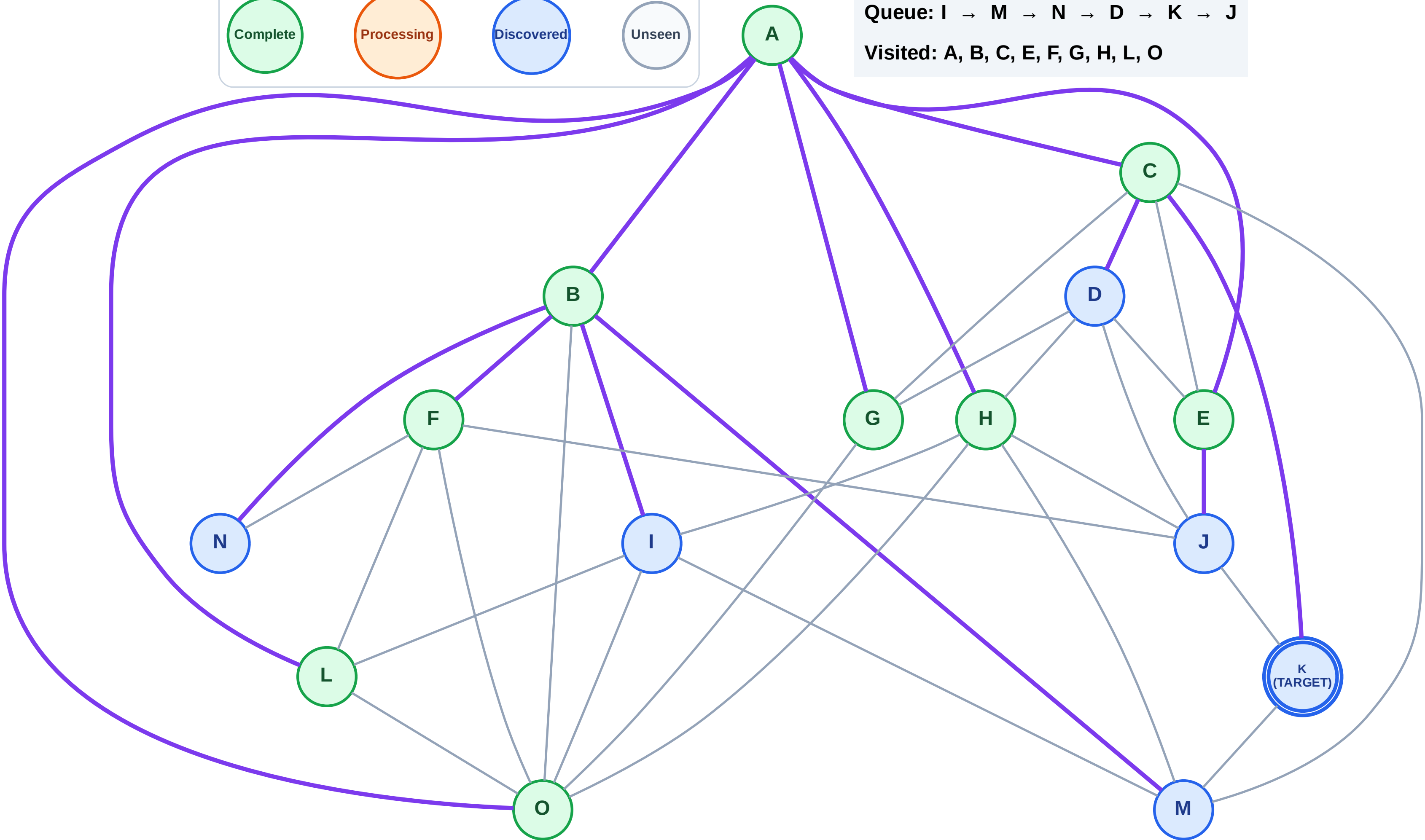
Step 19: No new neighbors from F

Legend



Queue: I → M → N → D → K → J

Visited: A, B, C, E, F, G, H, L, O



BFS Traversal

Step 20: Process node I

Legend

Complete

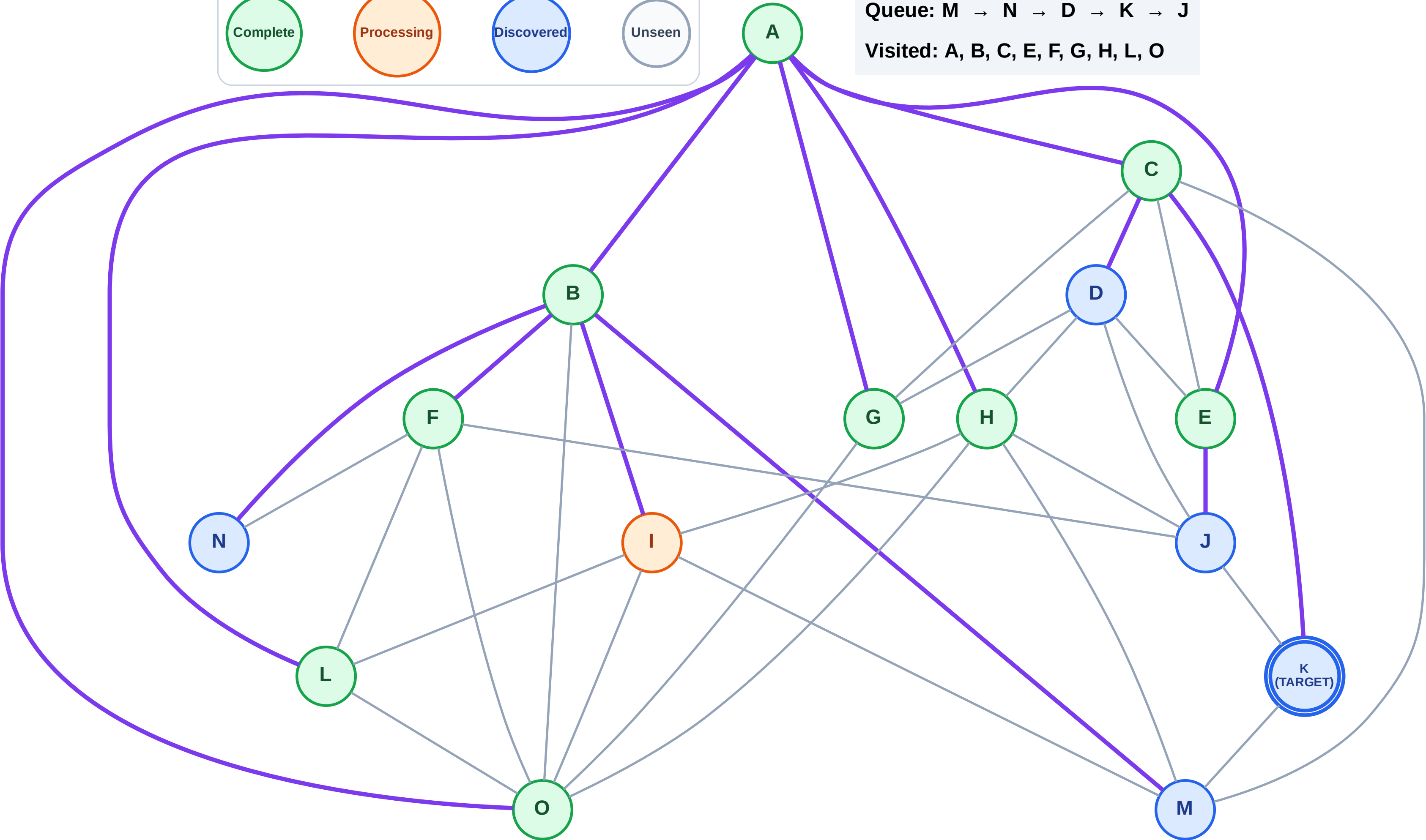
Processing

Discovered

Unseen

Queue: M → N → D → K → J

Visited: A, B, C, E, F, G, H, L, O



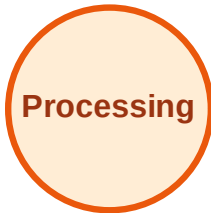
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



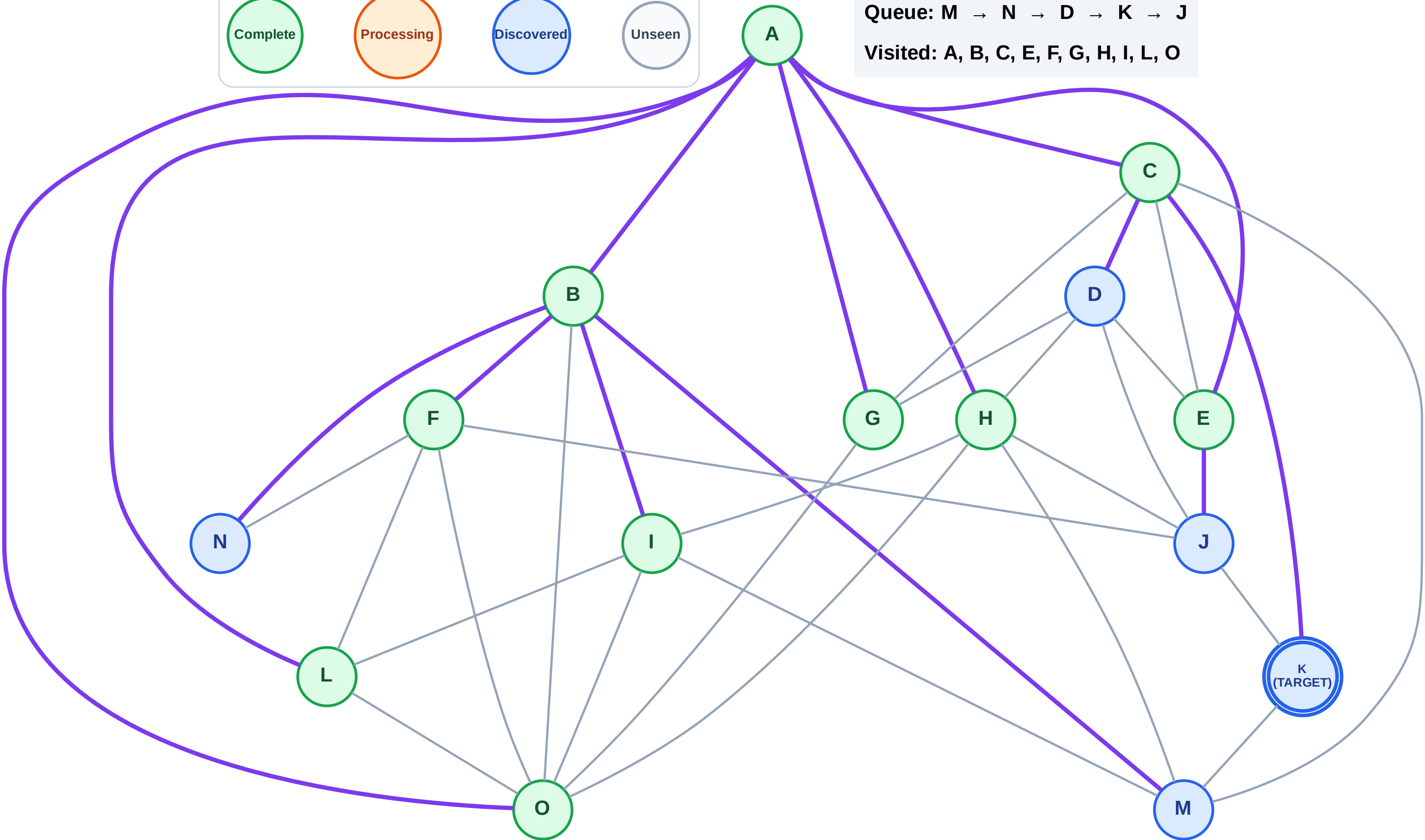
Discovered



Unseen

Queue: M → N → D → K → J

Visited: A, B, C, E, F, G, H, I, L, O



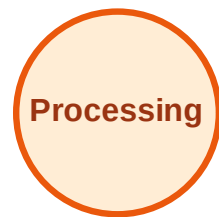
BFS Traversal

Step 22: Process node M

Legend



Complete



Processing



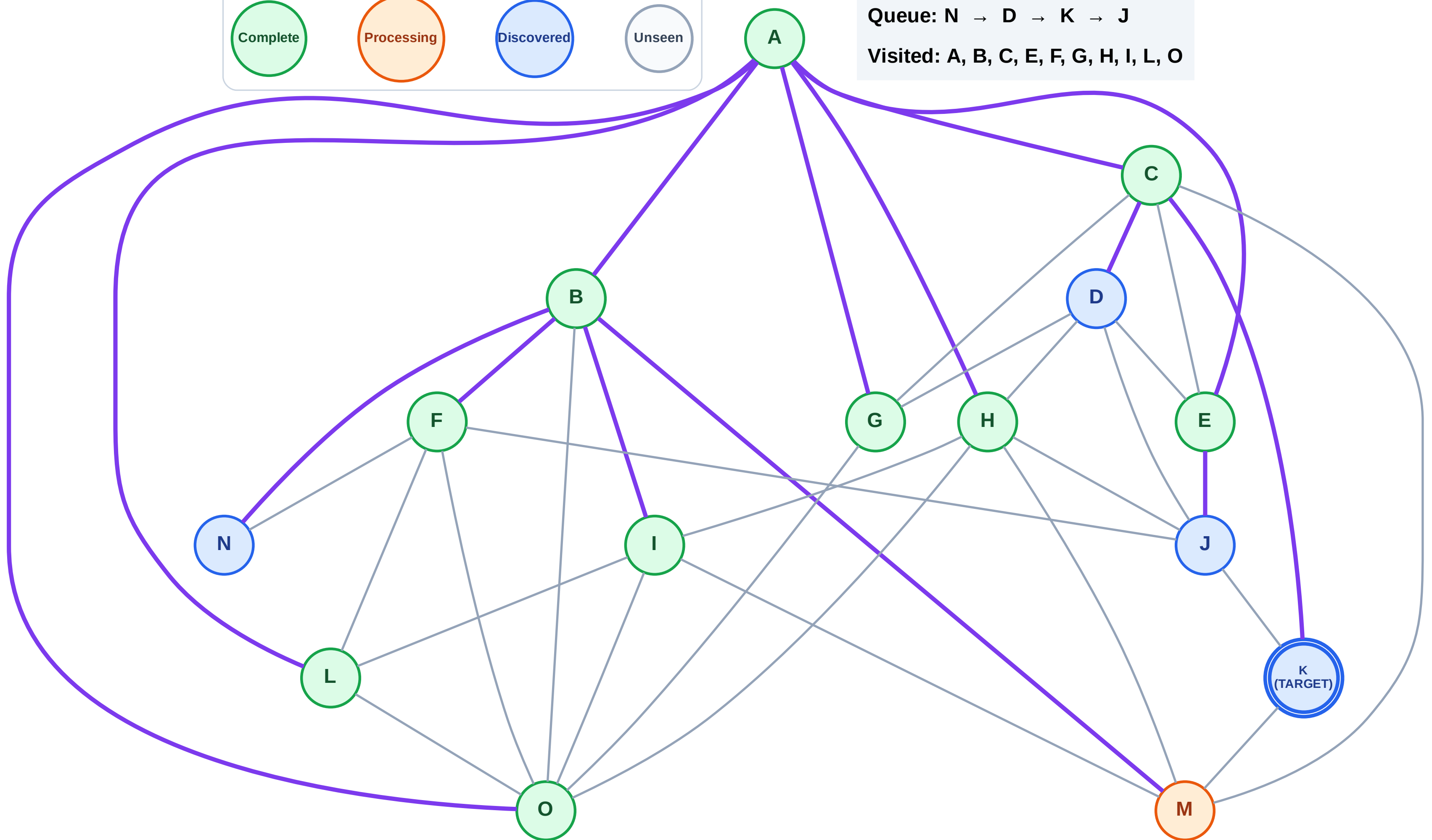
Discovered



Unseen

Queue: N → D → K → J

Visited: A, B, C, E, F, G, H, I, L, O



BFS Traversal

Step 23: No new neighbors from M

Legend

Complete

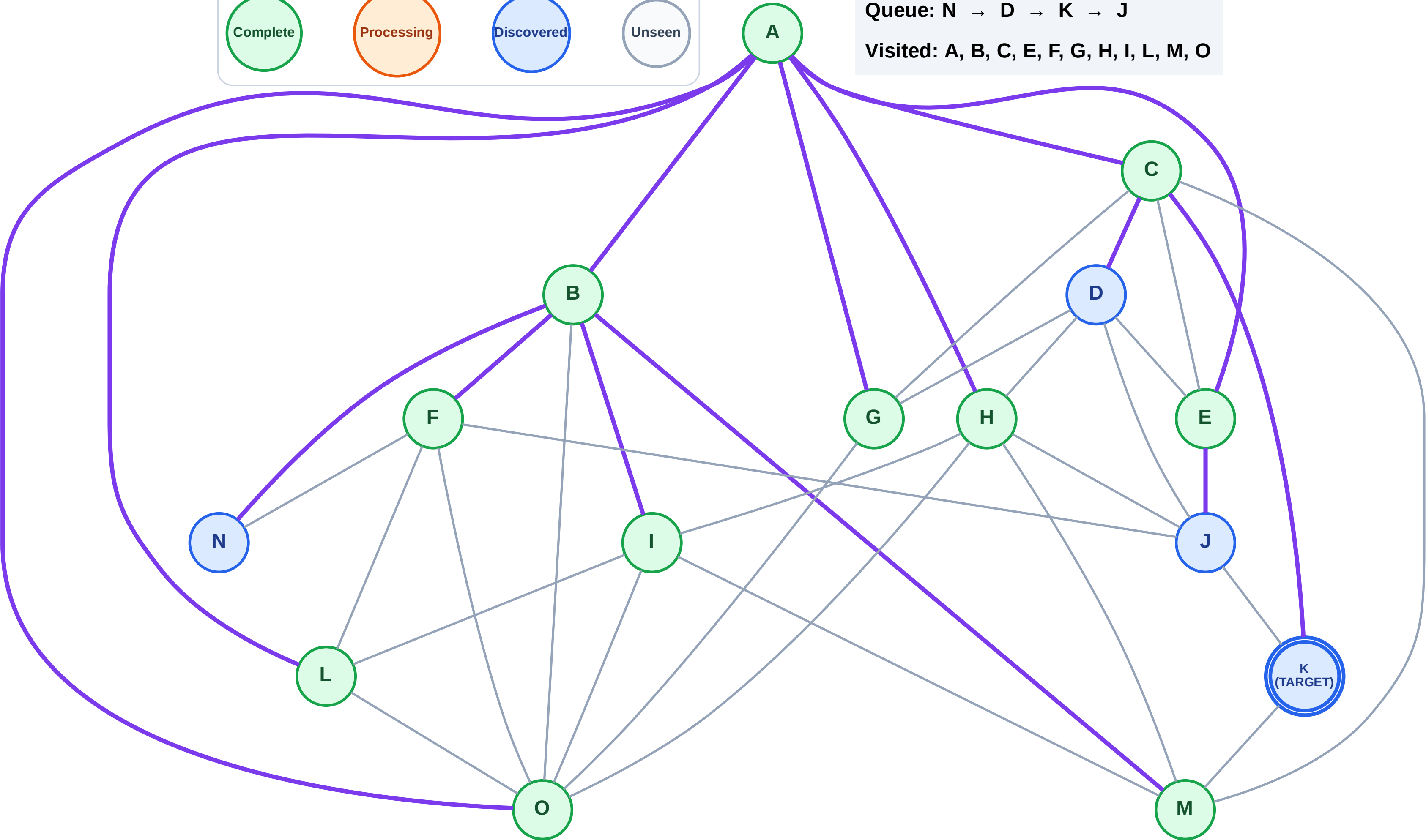
Processing

Discovered

Unseen

Queue: N → D → K → J

Visited: A, B, C, E, F, G, H, I, L, M, O



BFS Traversal

Step 24: Process node N

Legend

Complete

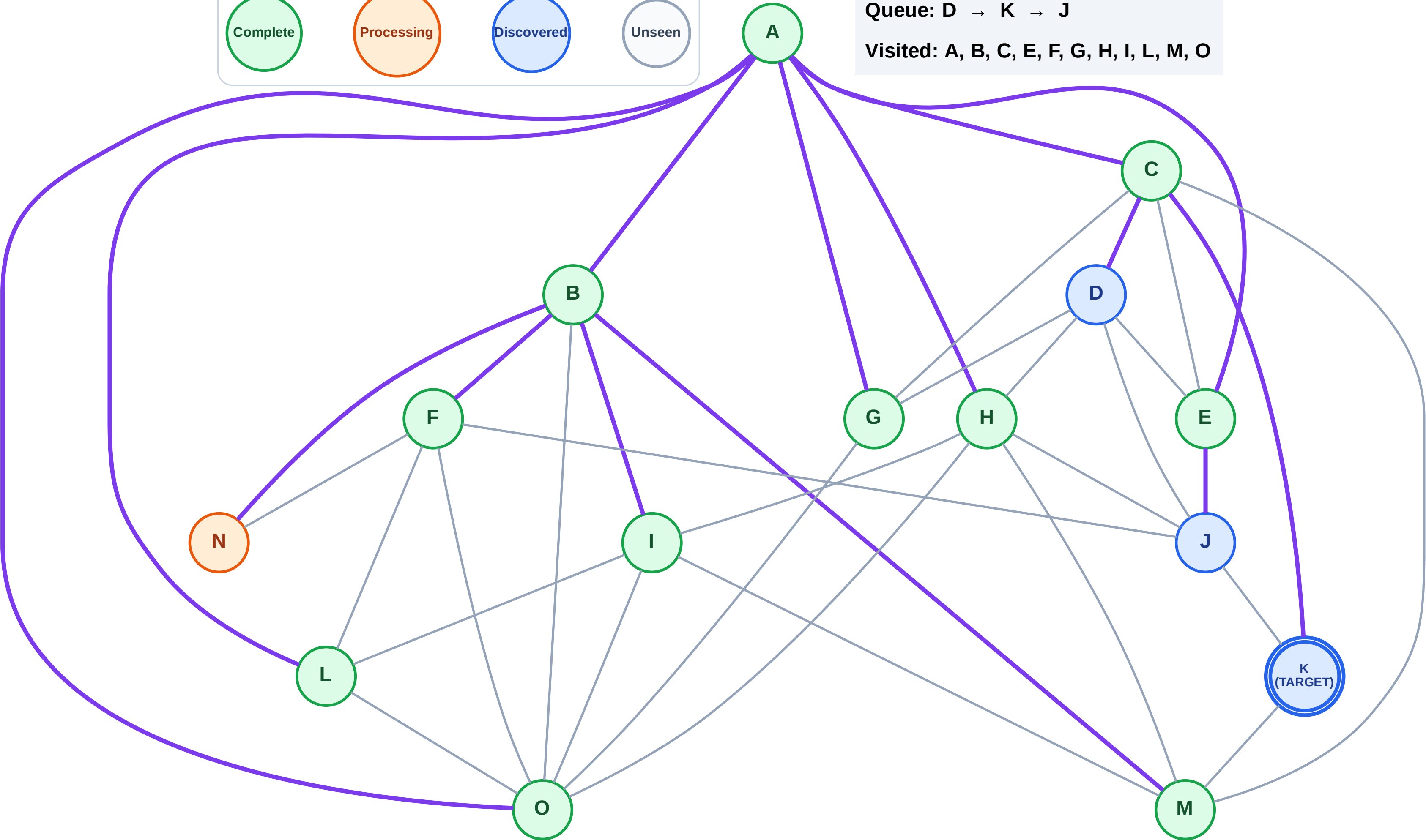
Processing

Discovered

Unseen

Queue: D → K → J

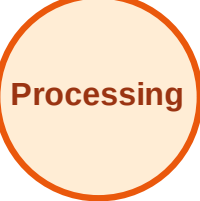
Visited: A, B, C, E, F, G, H, I, L, M, O



BFS Traversal

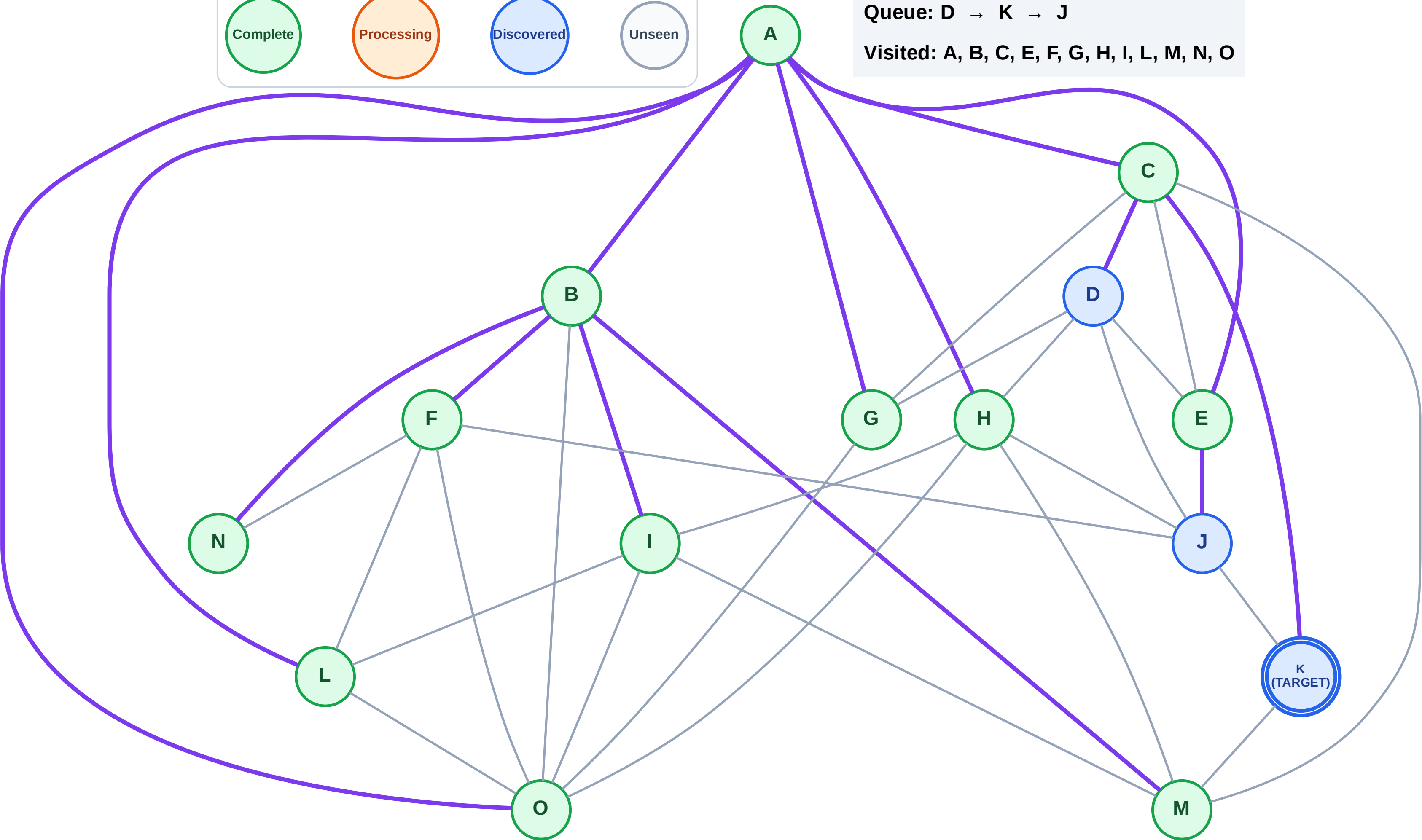
Step 25: No new neighbors from N

Legend



Queue: D → K → J

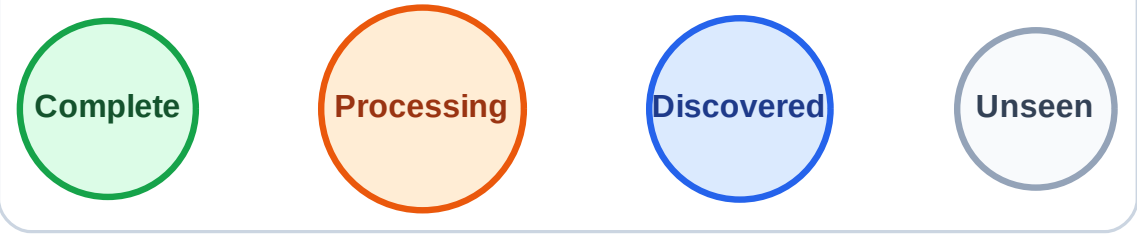
Visited: A, B, C, E, F, G, H, I, L, M, N, O



BFS Traversal

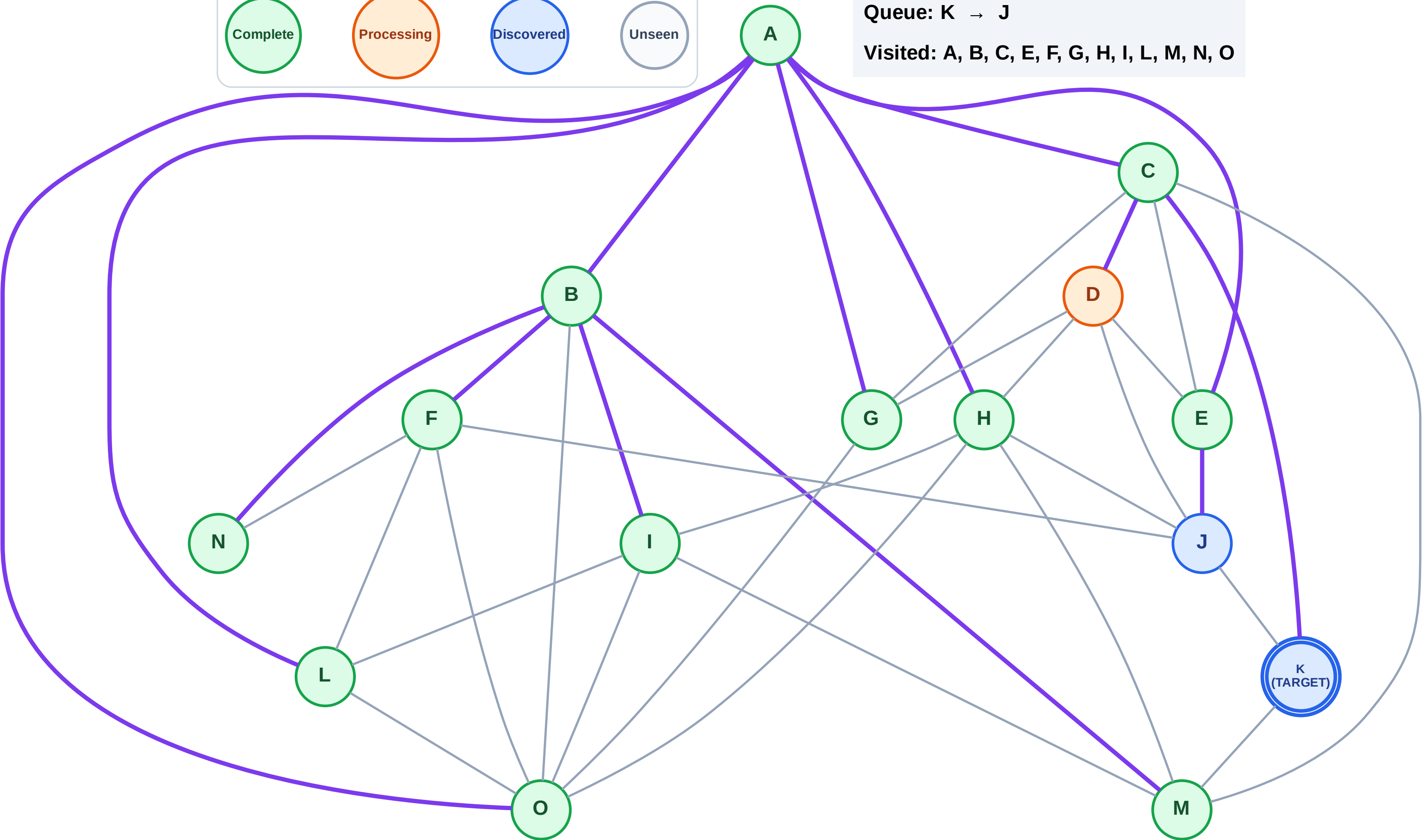
Step 26: Process node D

Legend



Queue: K → J

Visited: A, B, C, E, F, G, H, I, L, M, N, O



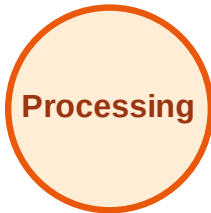
BFS Traversal

Step 27: No new neighbors from D

Legend



Complete



Processing



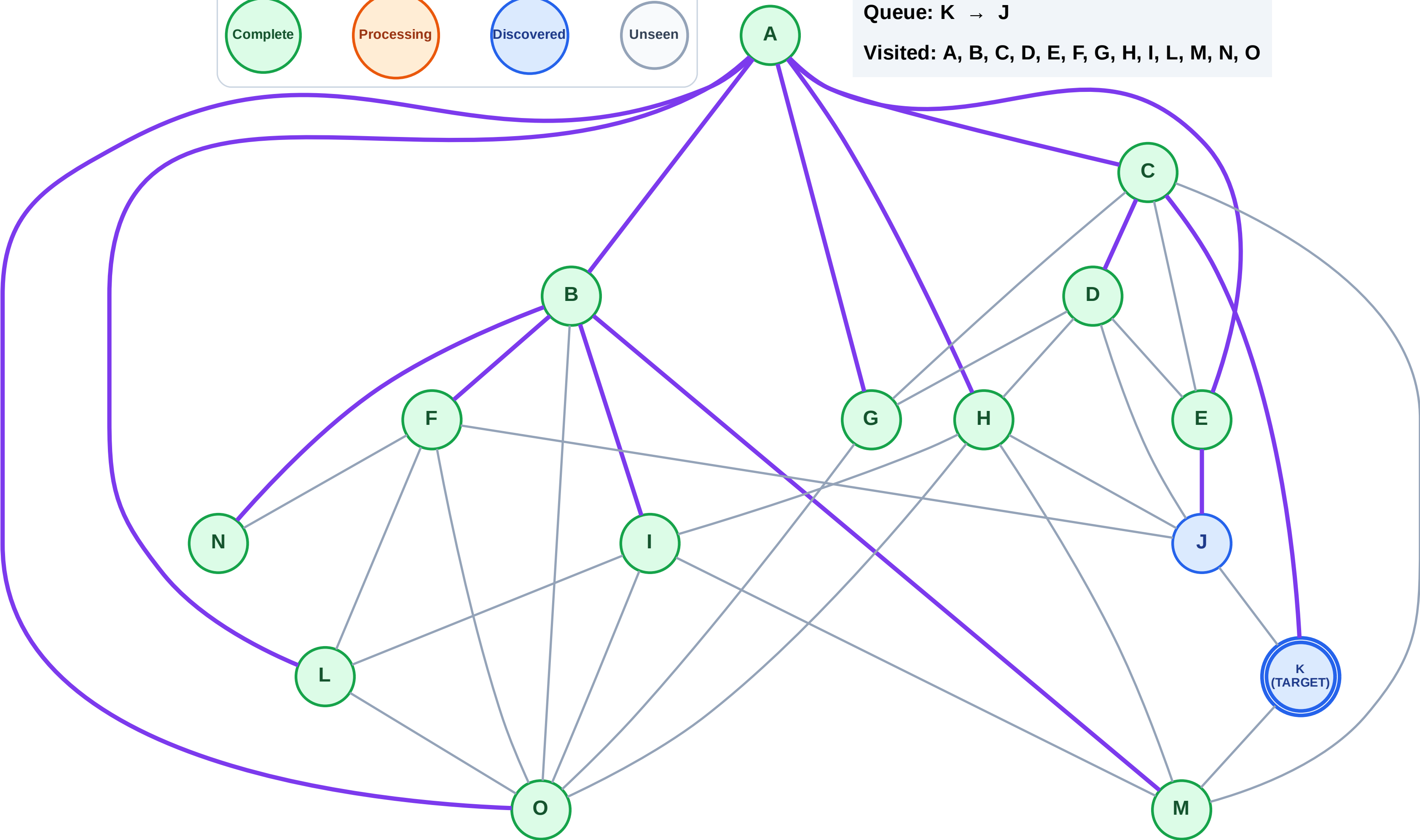
Discovered



Unseen

Queue: K → J

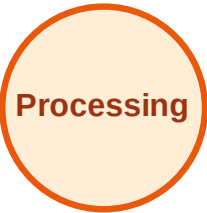
Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

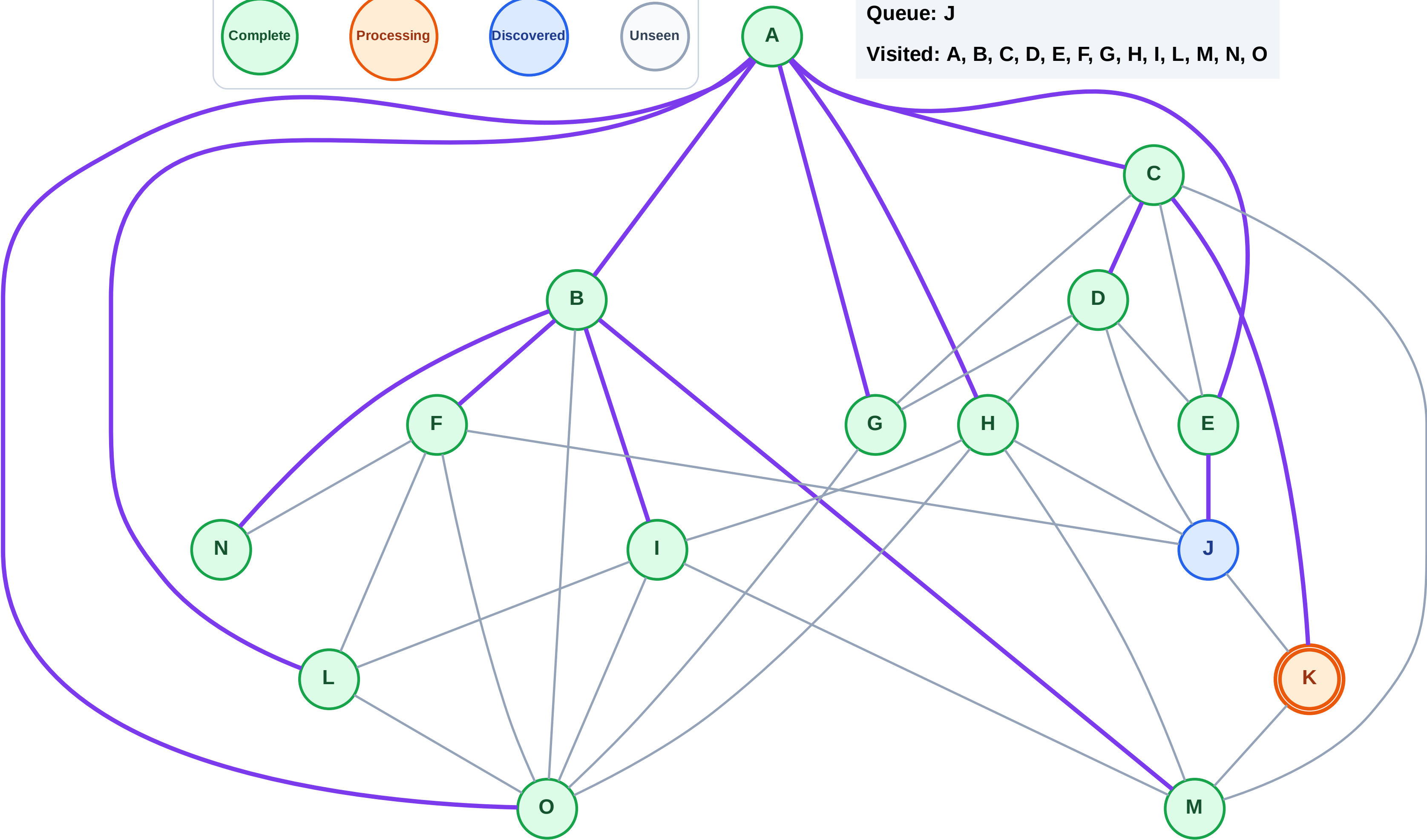
Step 28: Process node K

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I, L, M, N, O



BFS Traversal

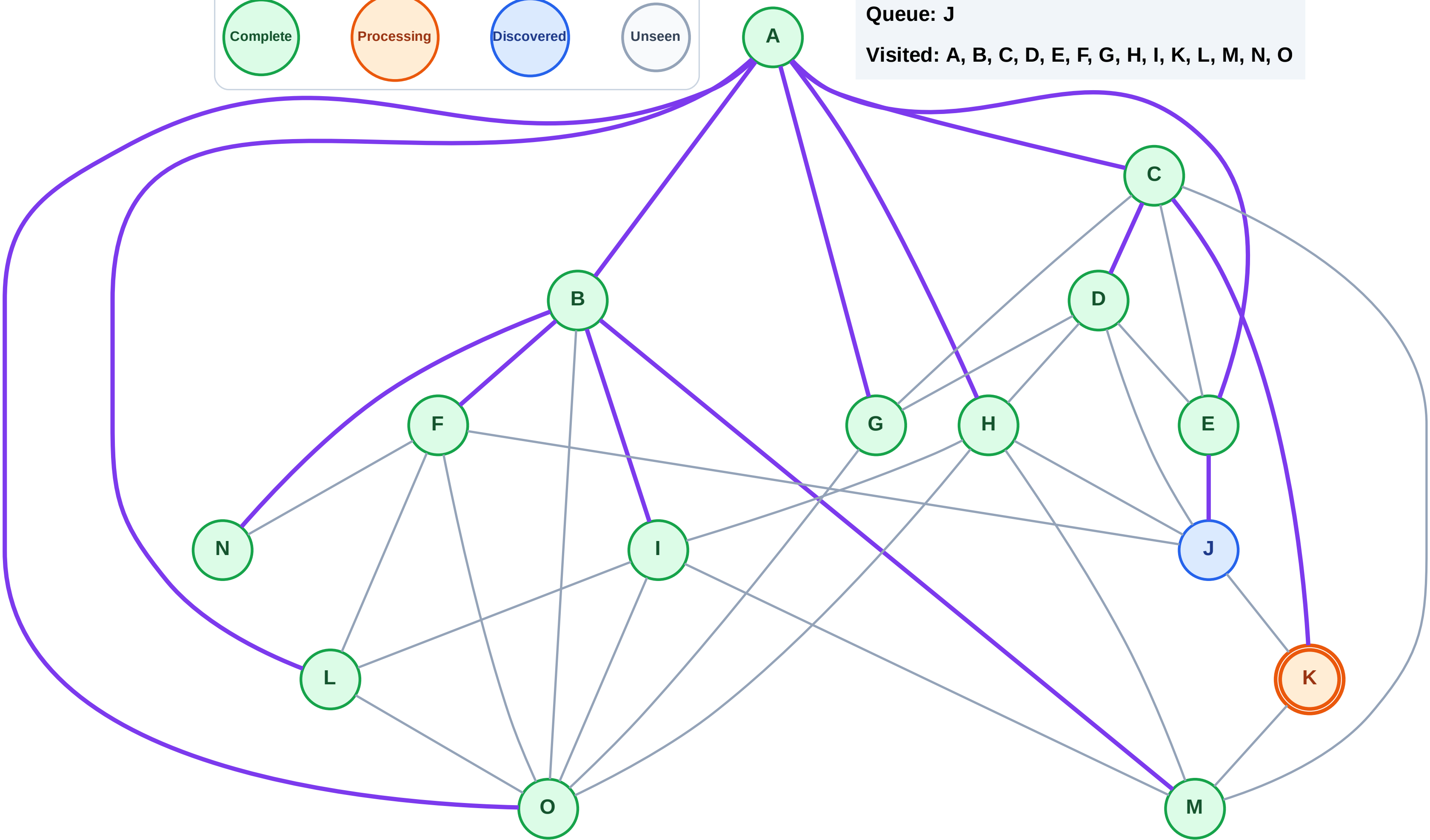
Step 29: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I, K, L, M, N, O



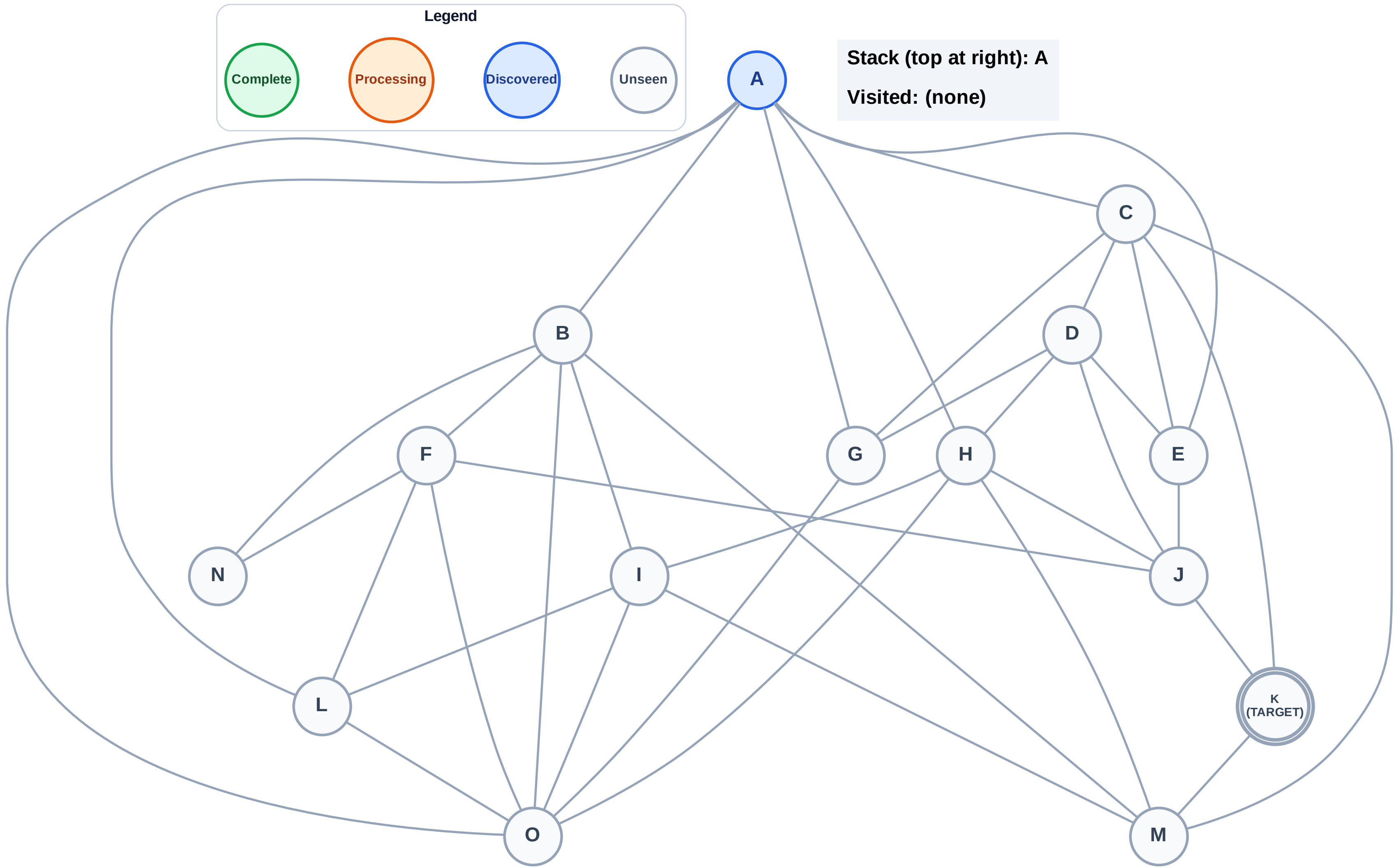
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

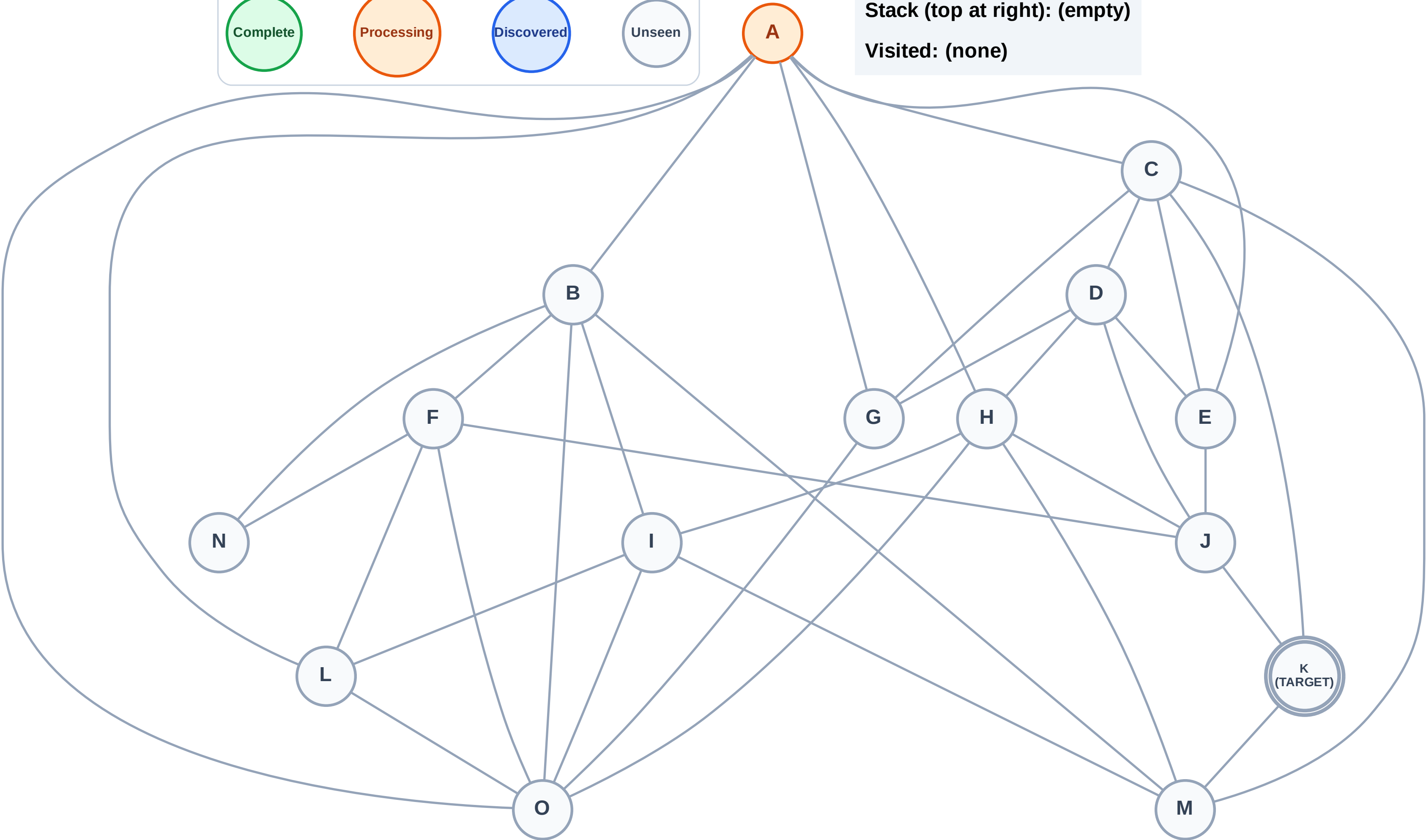
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, L, H, G, E, C, B from A

Legend

Complete

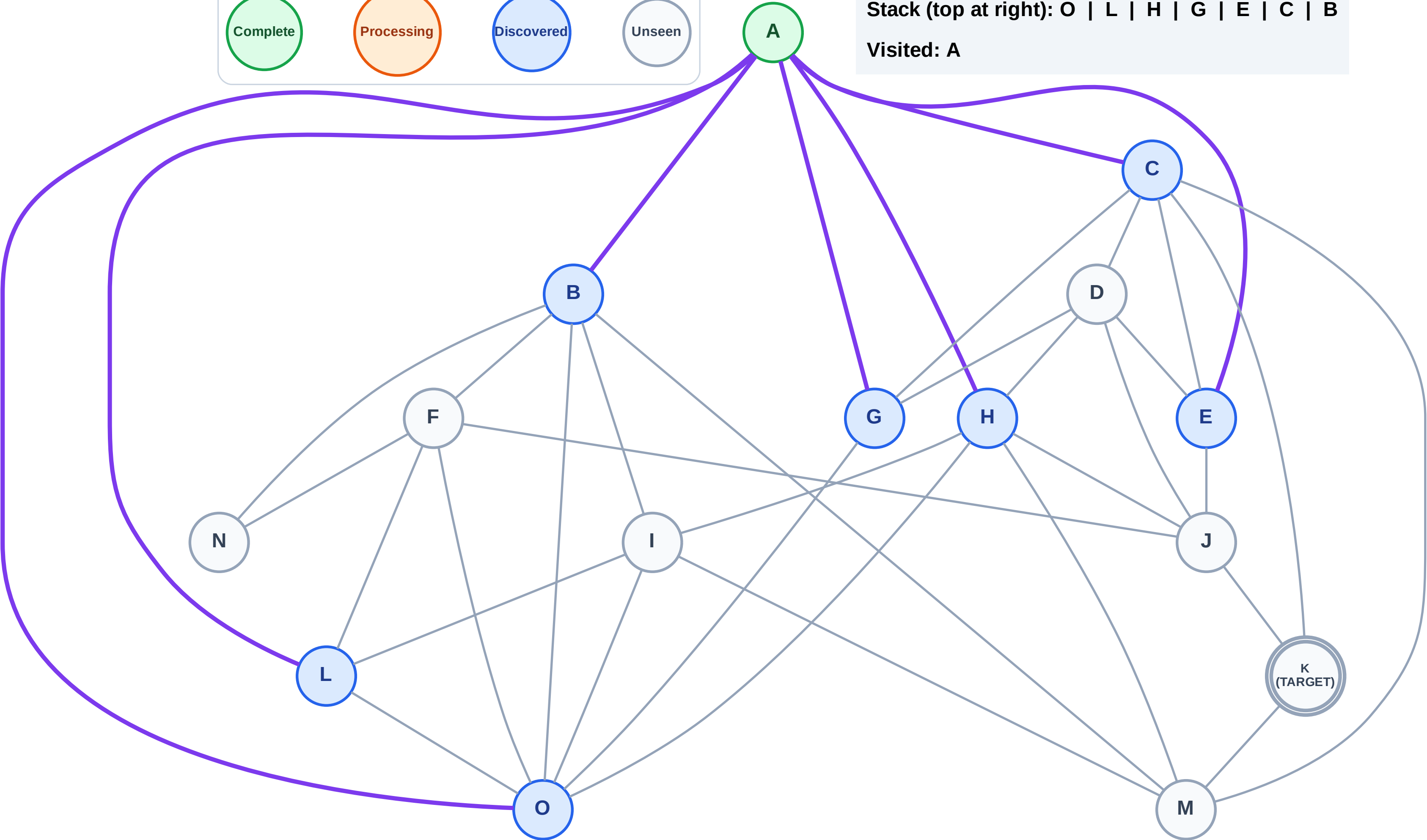
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

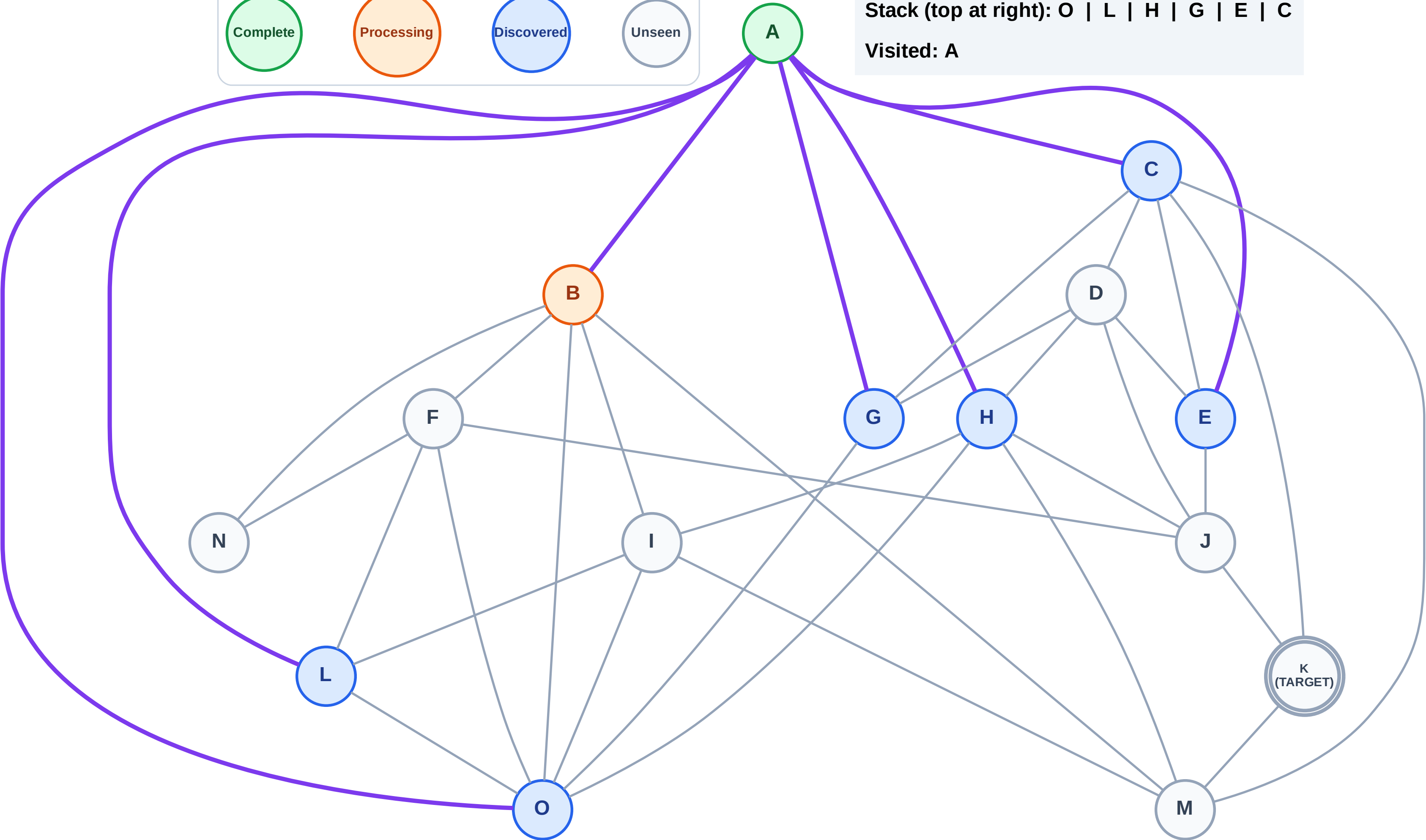
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C

Visited: A



DFS Traversal

Step 5: Discover N, M, I, F from B

Legend



Complete



Processing



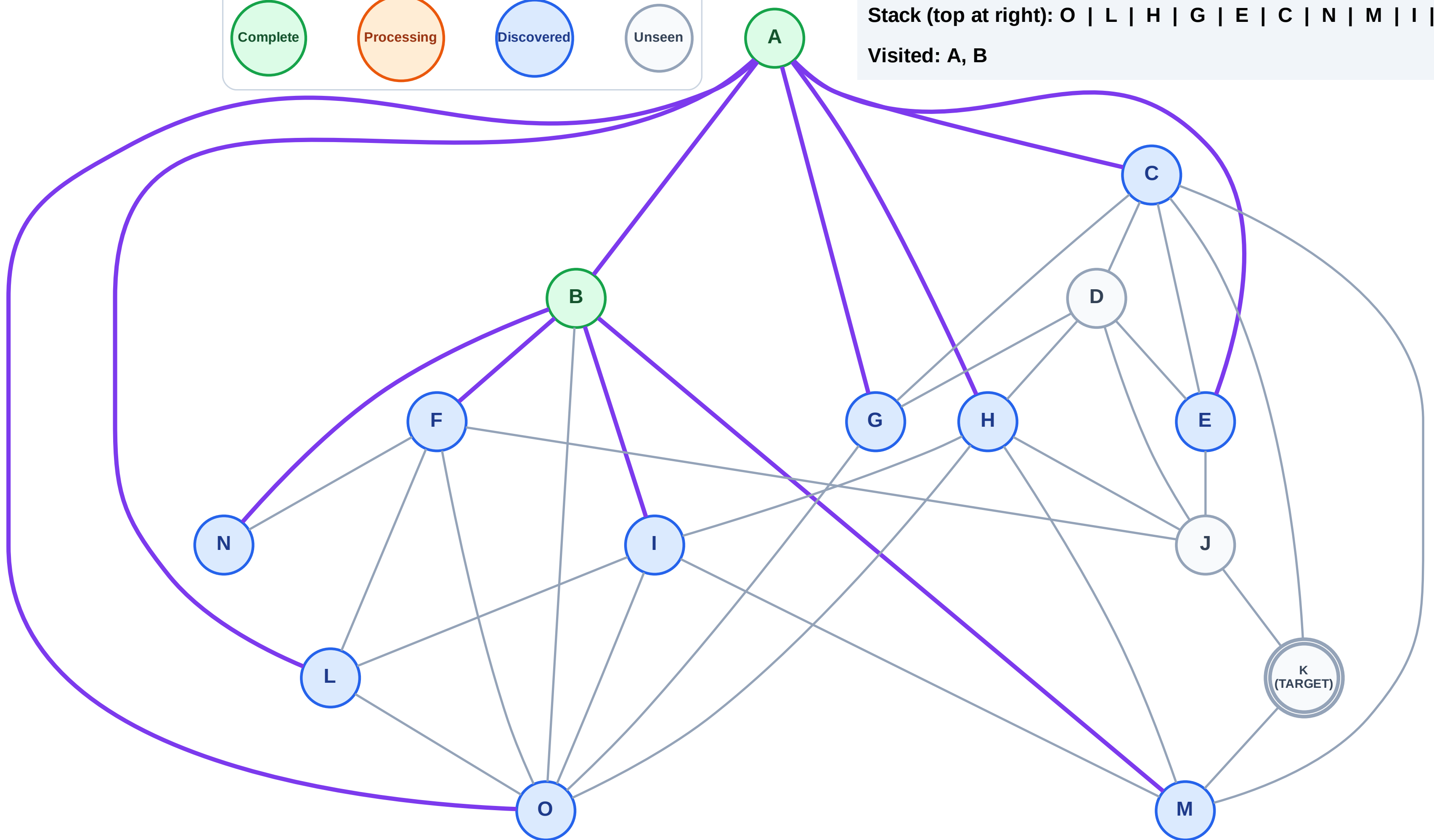
Discovered



Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

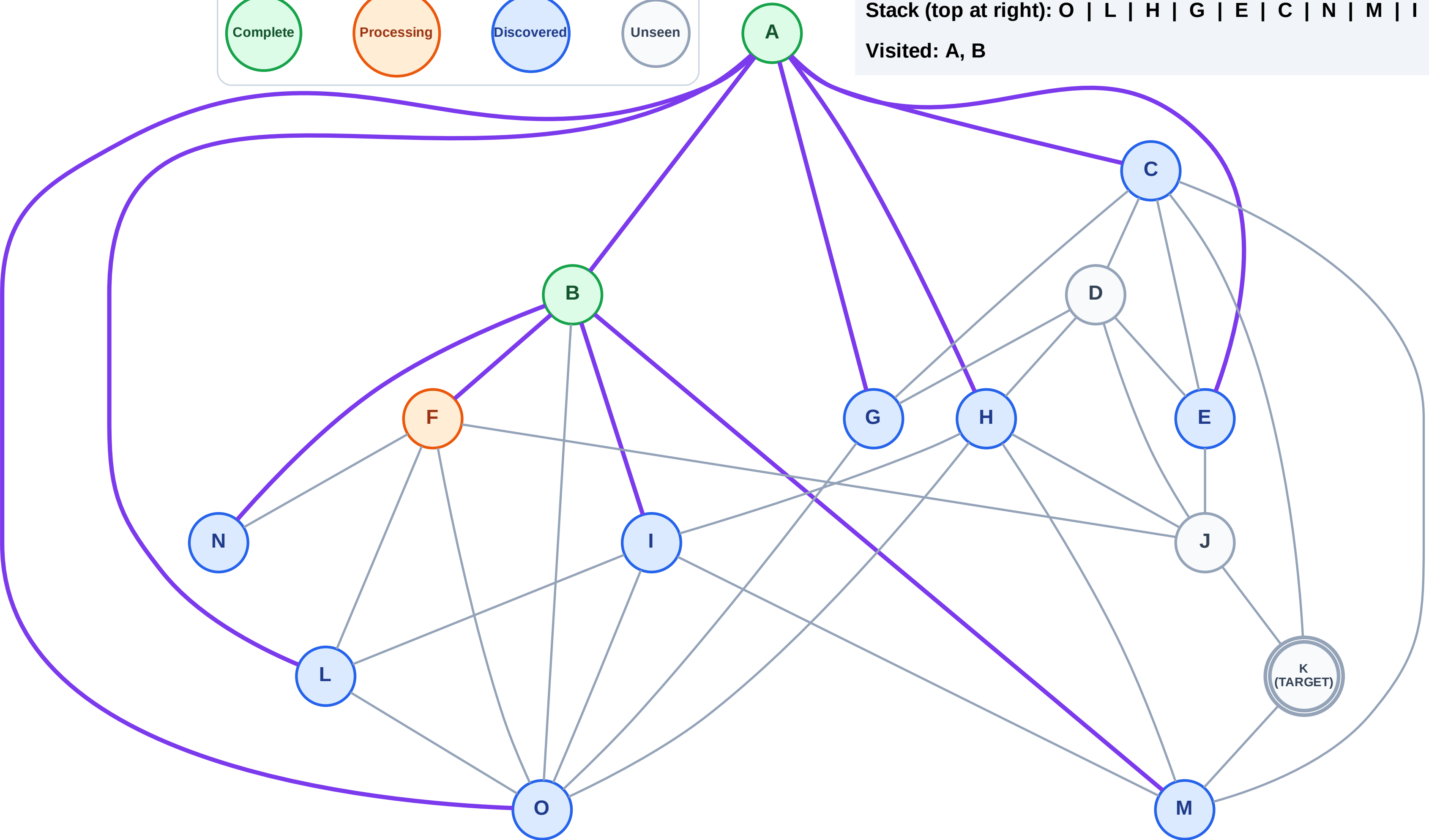
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B



DFS Traversal

Step 7: Discover J from F

Legend

Complete

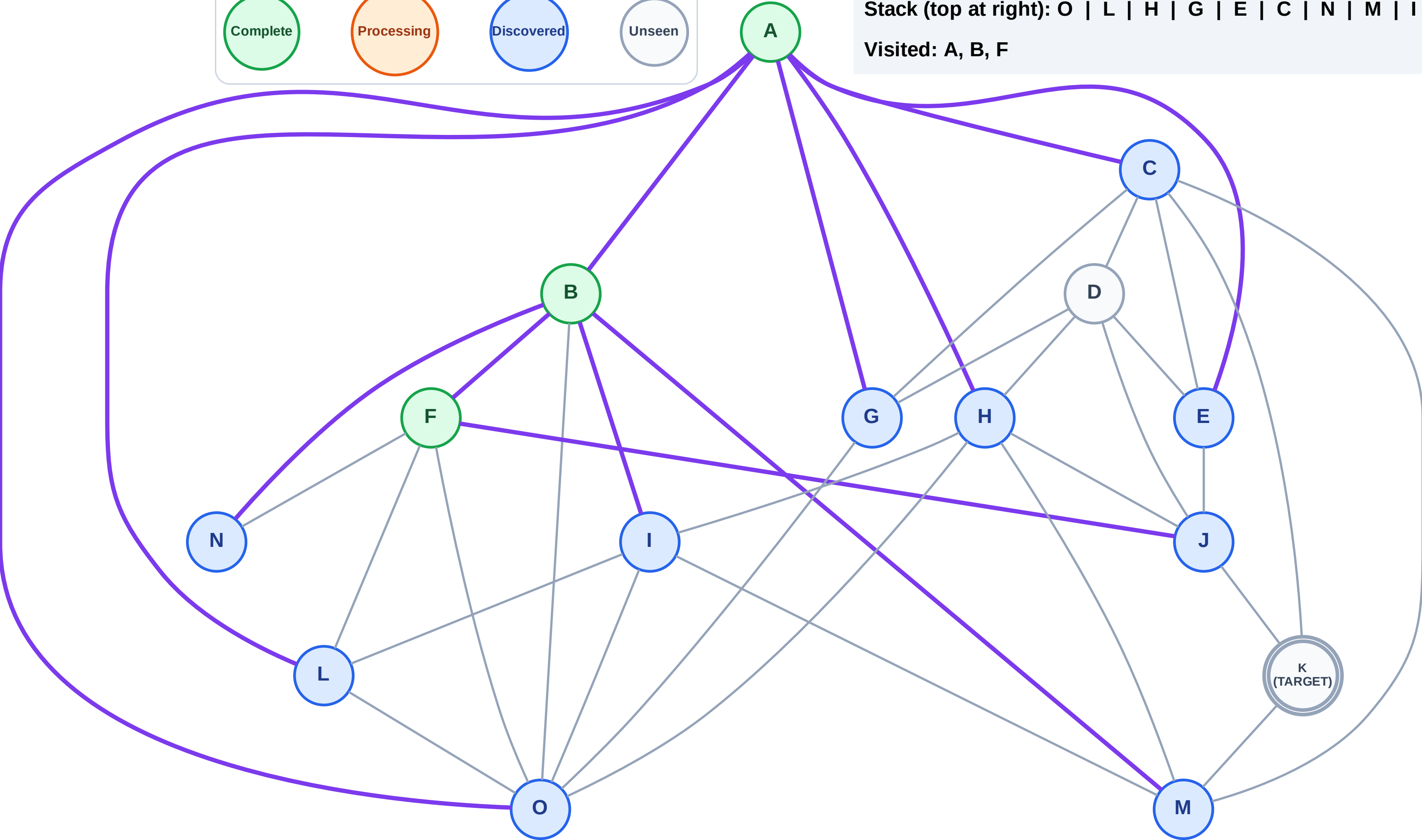
Processing

Discovered

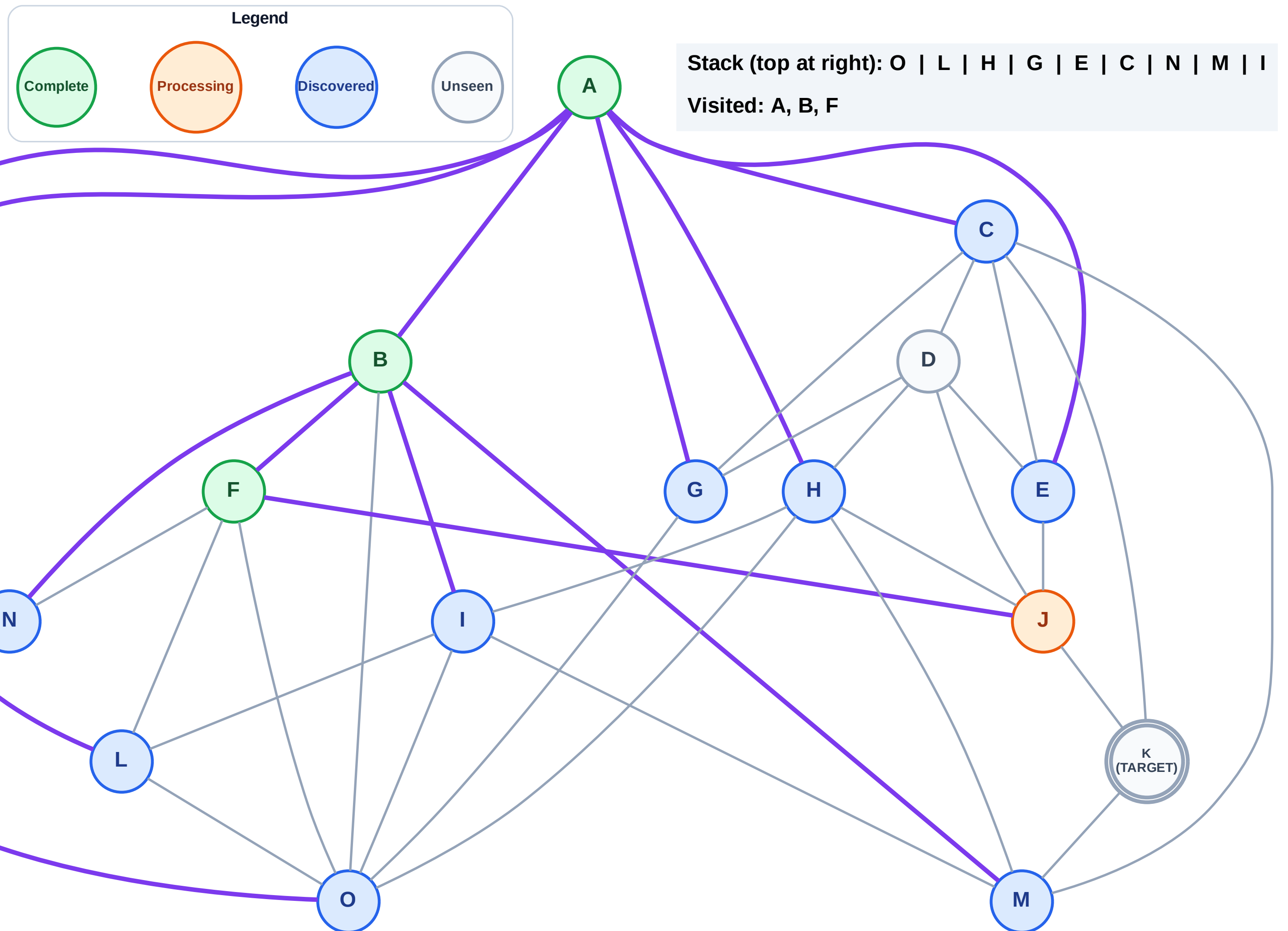
Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | J

Visited: A, B, F



Step 8: Process node J



DFS Traversal

Step 9: Discover K, D from J

Legend

Complete

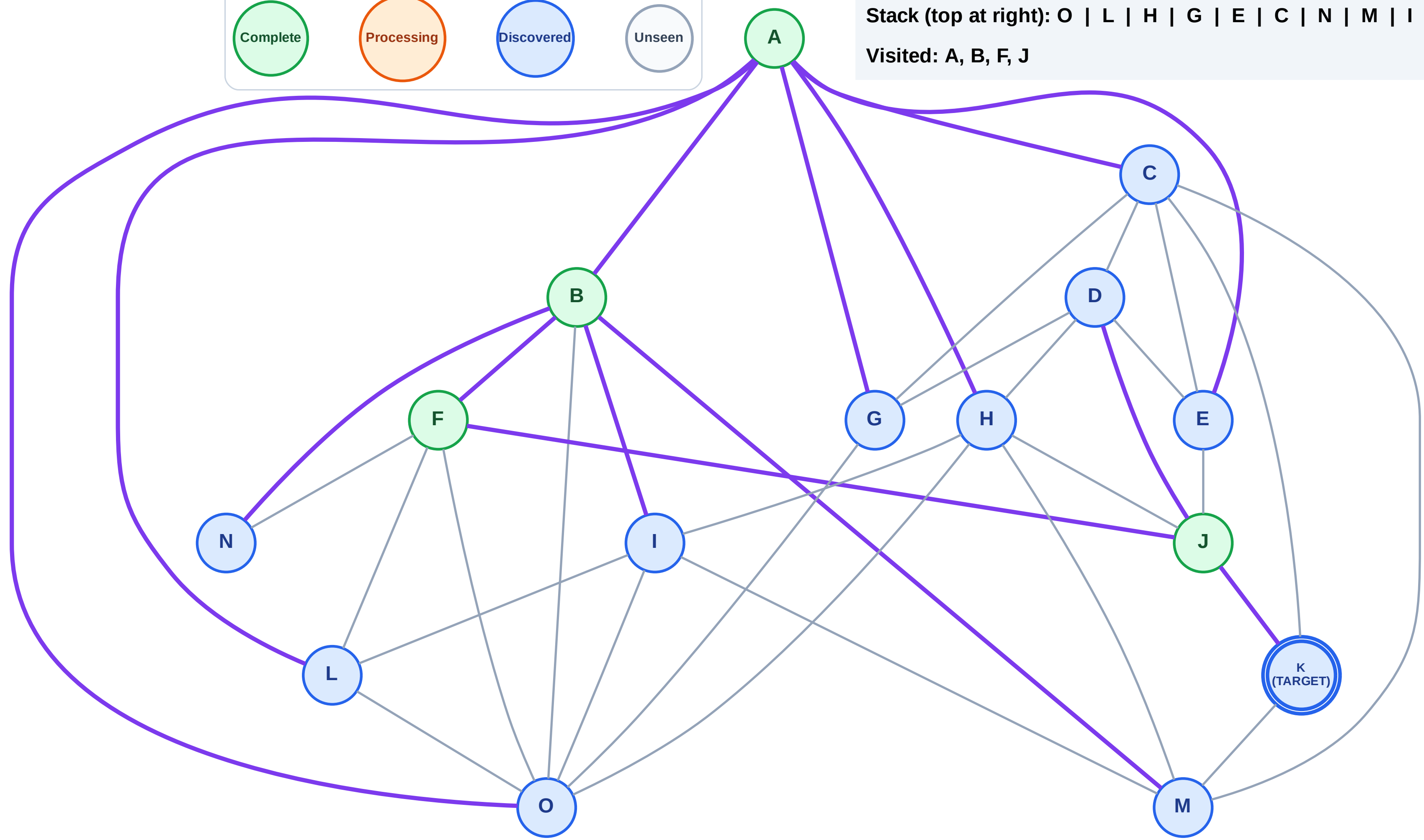
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | K | D

Visited: A, B, F, J



DFS Traversal

Step 10: Process node D

Legend

Complete

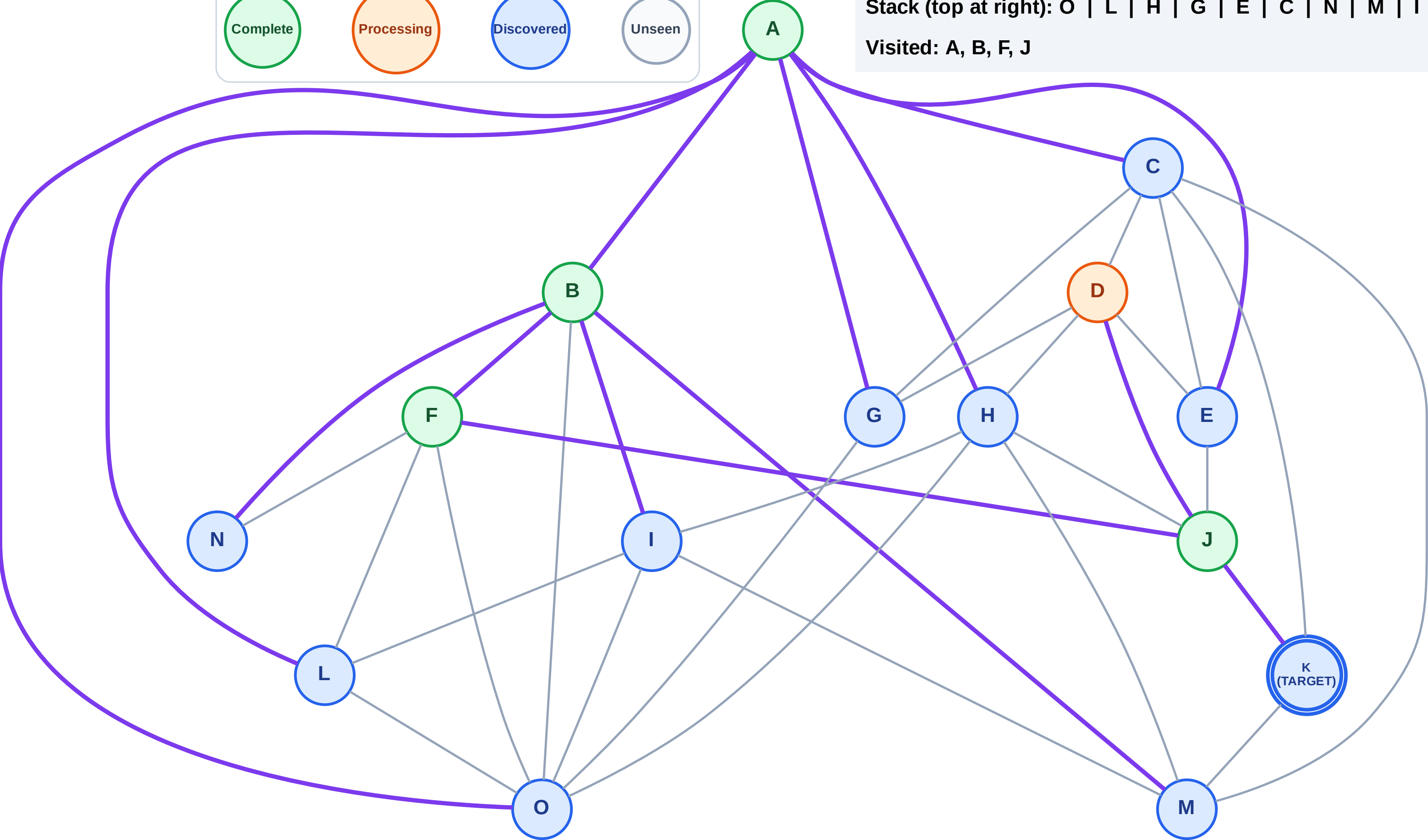
Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I | K

Visited: A, B, F, J



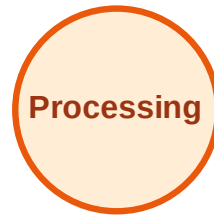
DFS Traversal

Step 11: No new neighbors from D

Legend



Complete



Processing



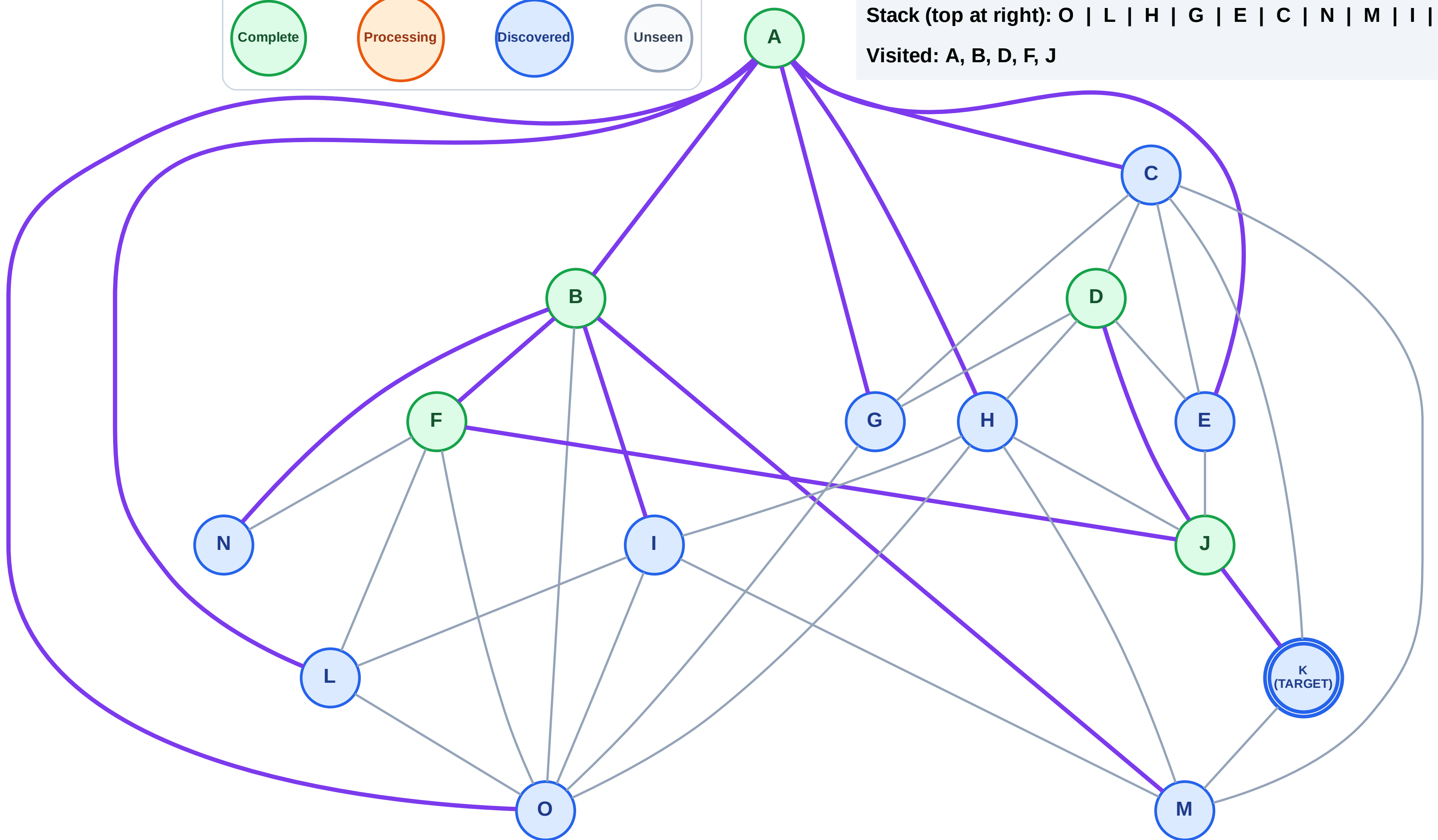
Discovered



Unseen

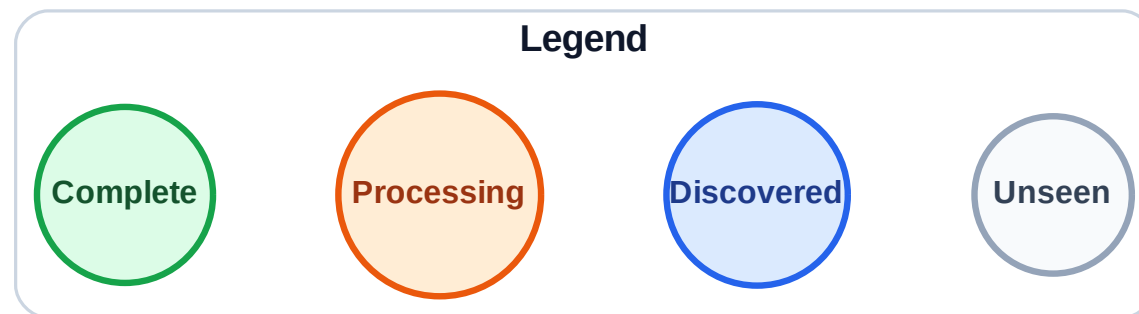
Stack (top at right): O | L | H | G | E | C | N | M | I | K

Visited: A, B, D, F, J



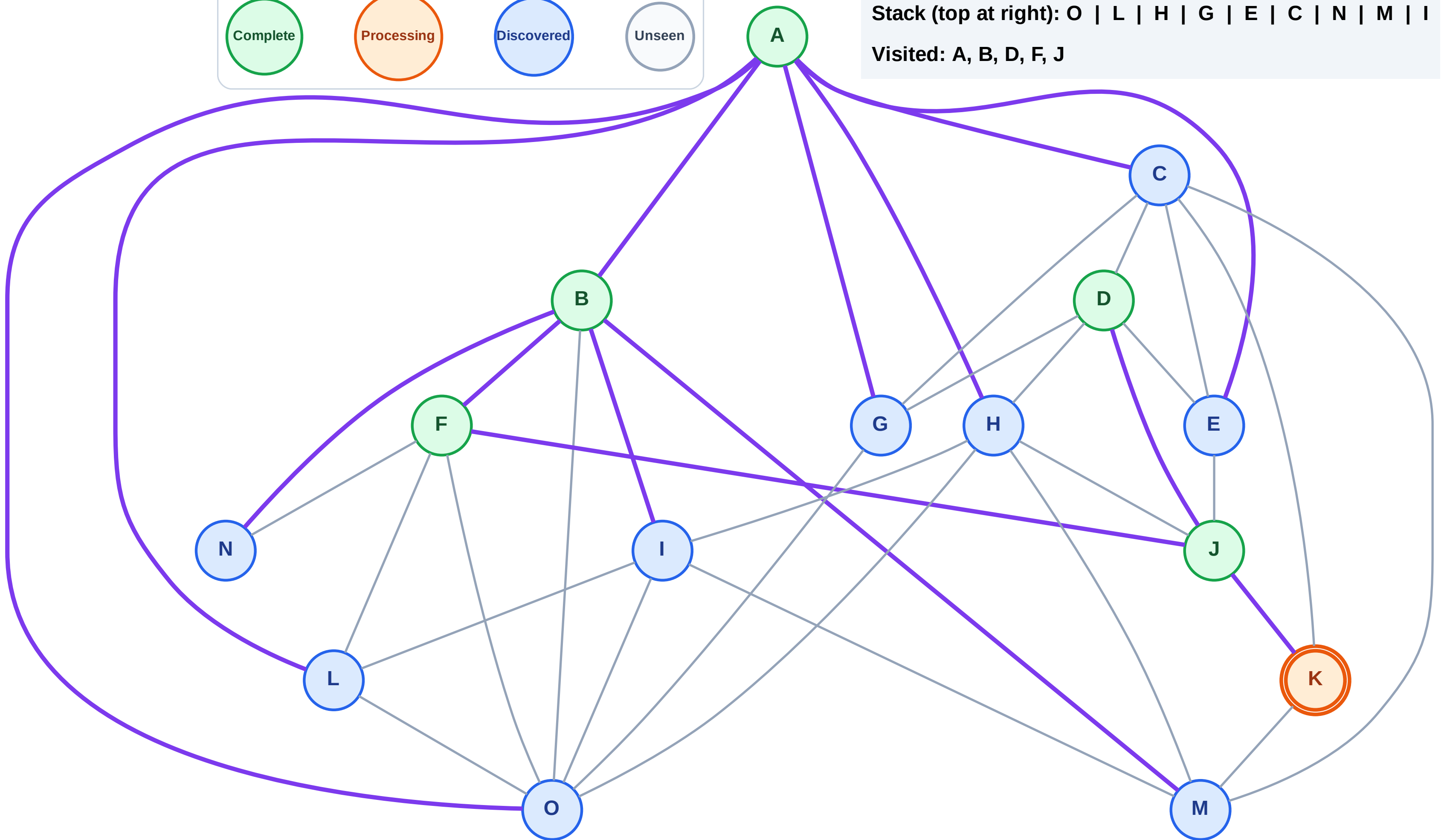
Step 12: Process node K

Step 12: Process node K



Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B, D, F, J



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (4 hops): A → B → F → J → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | L | H | G | E | C | N | M | I

Visited: A, B, D, F, J, K

